

Thin-walled structures

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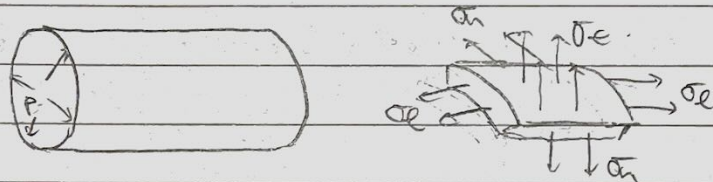
Thin-walled structures

- We can consider a structure to be thin-walled when the wall thickness $\ll$ other dimensions in the structure (for circular cylinders, $r/t \geq 20$)
- For a thin-walled structure, we can assume:
 - $\rightarrow$ through thickness stresses are zero, $\sigma_z = 0$
 - $\rightarrow$ the stress state is uniform through the section

Stresses

stresses due to internal pressure -

- Consider a circular cylinder under internal pressure p .



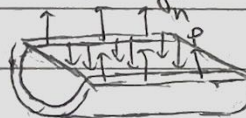
- Despite our assumption, the through-thickness stress cannot be constant. Since the internal pressure is p and the external pressure is 0, then

$$\sigma_r = -p \text{ on inner face}$$

$$\sigma_r = 0 \text{ on outer face}$$

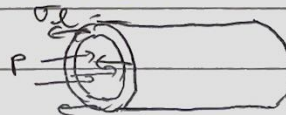
However, we will still assume $\sigma_z = 0$ everywhere ($|\sigma_r| \leq p \ll \sigma_\theta$)

- Consider the cut below to find the hoop stress σ_θ



$$p(2rL) = \sigma_\theta(2Lt) \rightarrow \boxed{\sigma_\theta = \frac{pr}{t}}$$

- Consider the cut below to find the longitudinal stress σ_z



$$p(\pi r^2) = \sigma_z(L\pi t) \rightarrow \boxed{\sigma_z = \frac{pr}{2t}}$$

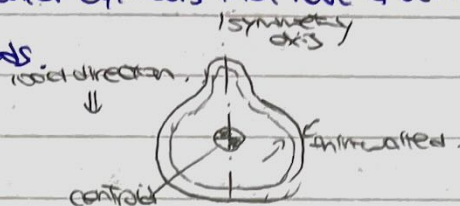
* σ_z depends on the support conditions - the above expression is for a closed end.

For pipelines w/ bellows expansion joint, $\sigma_z = 0$.

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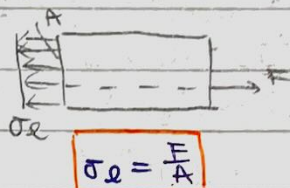
stresses due to axial forces, bending moments and shear forces.

- consider general, non-circular cylinders that have a vertical axis of symmetry, under the action of vertical loads



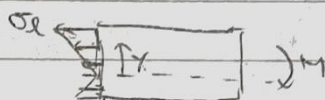
The axis of symmetry ensure that vertical loads and bending moments cause vertical deflections and curvatures in a vertical plane, but not sideways deflection and twist.

- Consider an applied axial force F .



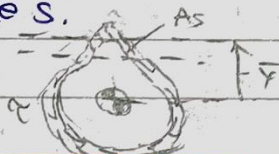
$$\sigma_x = \frac{F}{A}$$

- Consider an applied bending moment M .



$$\sigma_x = \frac{My}{I}$$

- Consider an applied shear force S .

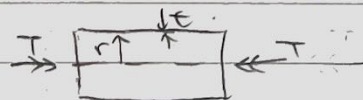


$$\tau(\text{at}) = q = \frac{SA_s y}{I}$$

* For τ_{max} , the cut should be $\perp$ to the wall to get τ_{max} .

Torsion.

- consider a uniform thin-walled circular cylinder subject to an applied torque T .



Due to axisymmetry, the shear stress τ must be constant.



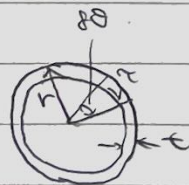
consider the eqn. of a small segment subject to an infinitesimal torque δT .

$$\delta T = \delta F \cdot r = \tau \delta A \cdot r = \tau (r d\theta) \cdot r$$

Integrating around the circle,

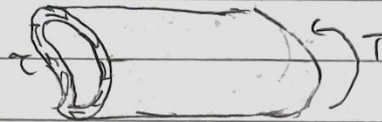
$$T = \int_0^{2\pi} \tau r^2 d\theta = \tau r^2 \cdot 2\pi t$$

$$\therefore \tau = \frac{T}{2\pi r^2 t}$$



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- Consider a general thin-walled cross-section cylinder subject to an applied torque T .

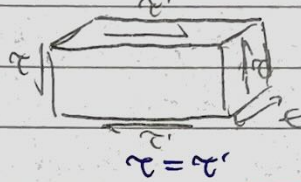


Due to the lack of symmetry, we can no longer assume the shear stress τ is constant around the cross-section $\rightarrow$ we need to find τ .

- Shear flow q is the shear force per unit wall length for thin-walled structures, given by

$$q = \tau t$$

- Just as complementary shear stress, there must always also be a complementary shear flow.

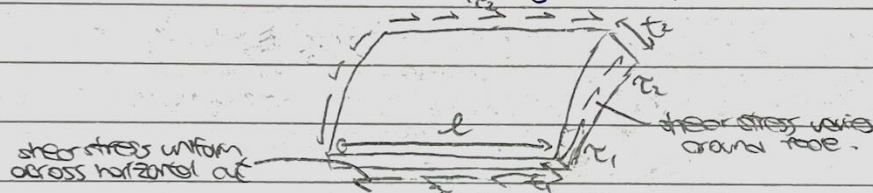


complementary shear:

$$\tau = \tau'$$

$$\therefore \tau t = \tau' t \rightarrow \boxed{q = q'}$$

- consider a section of thin-walled structure, of varying thickness, in a state of pure shear



N2L $\Rightarrow$:

$$\tau_1 A_1 = \tau_2 A_2$$

$$\tau_1 t_1 l = \tau_2 t_2 l$$

$$\boxed{\tau_1 = \tau_2}$$

$\rightarrow$ shear flow due to applied torque is constant around a thin-walled section.

- consider the eqn. of a small segment subject to an infinitesimal torque δT .

$$\delta T = \delta F \cdot r = \tau \delta A \cdot r = \tau (t \delta l) r$$

since we know shear flow $q = \tau t$ is constant around the section,

$$\delta T = q \cdot 2 \left(\frac{1}{2} r \delta l \right) = 2 q \delta A_e$$

where $\delta A_e = \frac{1}{2} r \delta l$ is the area enclosed by the small element.

Integrating around the section,

$$T = \oint 2 q \delta A_e$$

since q is constant,

$$\boxed{T = 2 q \oint A_e = 2 q A_e}$$

where A_e is the area enclosed by the section (to mid-thickness)

- therefore, for an arbitrary thin-walled closed section subject to an applied torque T ,

$$\boxed{q = \frac{T}{2 A_e}}$$

since $q = \tau t$,

$$\boxed{\tau = \frac{T}{2 A_e t}}$$

& the expressions above are not valid for open sections



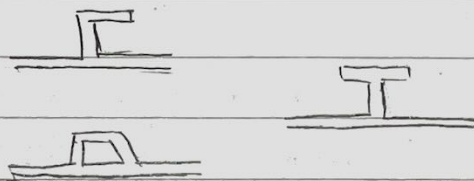
stiffeners

- Real thin-walled structures often have additional stiffeners to

→ act as a fail-safe device

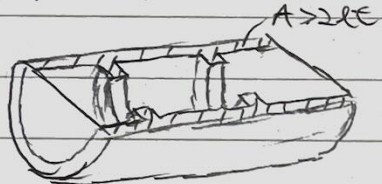
→ prevent local buckling of the walls

→ carry locally concentrated loads.



- For an internally pressurised circular cylinder w/ stiffeners.

① Transverse stiffeners



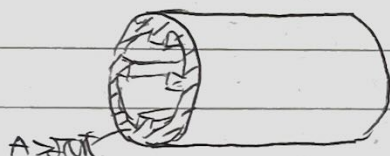
$$\bar{\sigma}_h A = 2rL \cdot p \quad A > 2Lt \quad (\text{due to stiffeners})$$

$$\therefore \bar{\sigma}_h < \frac{pr}{t} \quad [\bar{\sigma}_h \text{ reduced}]$$

* If we cut a free body that excluded stiffeners, $\bar{\sigma}_h = \frac{pr}{t}$

→ σ_h varies along the section, $\bar{\sigma}_h = \frac{pr}{t}$ is a conservative estimate.

② Longitudinal stiffeners.



$$\bar{\sigma}_x A = \pi r t \cdot p \quad A > 2\pi r t \quad (\text{due to stiffeners})$$

$$\therefore \bar{\sigma}_x < \frac{pr}{t} \quad [\bar{\sigma}_x \text{ reduced}]$$

* σ_x does not vary around the section, $\bar{\sigma}_x = \frac{pr}{t}$ is a conservative estimate.

- Longitudinal stiffeners will increase the cross-sectional area of the structure, hence reduce the stress due to axial loads.

$$\text{decrease} \rightarrow \sigma_x = \frac{F}{A} \leftarrow \text{increase}$$

- Longitudinal stiffeners also increase the second moment of area for the section, and hence reduce the stress due to bending.

$$\text{decrease} \rightarrow \sigma_x = \frac{My}{I} \leftarrow \text{increase}$$

- Longitudinal stiffeners do not increase the shear stress due to shear loading as the increase in I is generally matched by an increase in $As\bar{y}$.

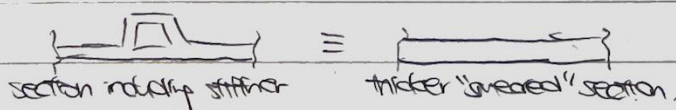
$$\text{roughly const.} \rightarrow q = \frac{SA\bar{y}}{I} \leftarrow \text{increase}$$

- Longitudinal stiffeners do not increase the area enclosed by a section, so the shear flow is unaffected.

$$\text{const.} \rightarrow q = \frac{T}{2Ae} \checkmark \text{const.}$$

* The local shear stress (due to shear force/torque) may decrease where $t = t_{\max} > t_{\min}$ since $\tau = \frac{q}{t}$.

- To analyse thin-walled structures w/ stiffeners, we consider thicker, "smeared" sections.



* When calculating the max. shear force (due to shear force/torque), we use the actual wall thickness $t_{\min}$. (not the smeared wall thickness)

- A transverse diaphragm could prevent buckling by reducing the effective length.

$$\text{increase} \rightarrow P_E = \frac{\pi^2 EI}{L_e^2} \leftarrow \text{decrease}$$

stress-strain relationships

Linear-elastic material model

- The linear-elastic material model has the following stress-strain characteristics

↳ linear: stress-strain curve is a straight line

↳ elastic: deformation disappears when load is removed.

- We can also assume that a linear-elastic material is

↳ homogeneous: doesn't vary w/ position

↳ isotropic: doesn't vary w/ direction.

↳ time independent: no creep

Normal strain

- The strain due to normal stress in three $\perp$ directions can be found by superposition

$$\epsilon_{xx} = \frac{1}{E} \sigma_{xx} - \frac{\nu}{E} \sigma_{yy} - \frac{\nu}{E} \sigma_{zz}$$

- A body subjected to a temp. rise undergoes a uniform strain in all directions and no shear strain. The normal strain is given by.

$$\epsilon_{xx} = \epsilon_{yy} = \epsilon_{zz} = \alpha \Delta T$$

where α is the coefficient of linear expansion.

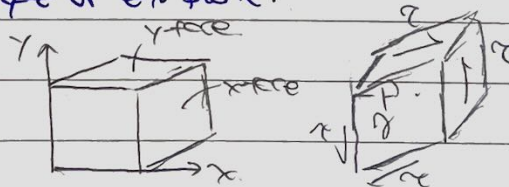
- The total strain due to normal stresses and temp. change is therefore.

$$\epsilon_{xx} = \frac{1}{E} (\sigma_{xx} - \nu \sigma_{yy} - \nu \sigma_{zz}) + \alpha \Delta T$$

* In an isotropic material, there is never any shear strain due to normal stress + temp. change alone.

Shear strain

- A shear stress on the x -face acting in the y -direction will cause a shear strain in the xy plane, but no shear strains on the yz or zx plane.



$$\sigma_{xy} = \frac{1}{G} \tau_{xy}$$

* In an isotropic material, a shear stress will only cause shear strain in its own plane and no normal strain in any direction

change in dimension and strain.

- The change in dimension can be related to strain as follows,

Length $\frac{\Delta l}{l} = \epsilon_l$

Circumference $\frac{\Delta C}{C} = \epsilon_h$

Radius $\frac{\Delta r}{r} = \frac{1}{2} \frac{\Delta C}{C} = \frac{\Delta C}{C} = \epsilon_h$ [Not ϵ_r !]

Thickness $\frac{\Delta t}{t} = \epsilon_t$

- The volumetric strain is defined as

$$\text{volumetric strain} = \frac{\Delta V}{V}$$

the expression for volumetric strain depends on the expression for volume.

e.g: $V = \pi r^2 l \rightarrow \frac{\Delta V}{V} = 2 \frac{\Delta r}{r} + \frac{\Delta l}{l} = 2\epsilon_h + \epsilon_l$

$V = 2\pi r t l \rightarrow \frac{\Delta V}{V} = \frac{\Delta r}{r} + \frac{\Delta t}{t} + \frac{\Delta l}{l} = \epsilon_h + \epsilon_t + \epsilon_l$

Torsional rigidity

Terms and definitions

- The twist per unit length ϕ is given by

$$\phi = \frac{\theta}{l}$$

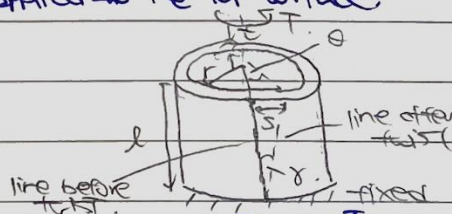
- The torsional rigidity is defined as

$$\text{torsional rigidity} = \frac{T}{\phi} = GJ$$

where G is the shear modulus and J is the torsion constant.

Uniform thin-walled circular section

- consider a thin-walled circular cylinder of thickness t , radius r , length l , whose base is fixed, while a torque T is applied to the top surface.



Equilibrium

$$\tau = \frac{T}{2\pi r t l} = \frac{T}{2\pi r^2 t l}$$

Compatibility

$$s = r\theta = l\phi$$

Material law

$$\tau = G\gamma$$

$$\therefore \frac{T}{2\pi r^2 t l} = G \left(\frac{\phi}{l} \right)$$

$$\frac{T}{\phi} = G(2\pi r^2 t l)$$

- The shear stress τ (constant in the section) is given by

$$\tau = \frac{T}{2\pi r^2 t l} = \frac{G(2\pi r^2 t l) \phi}{2\pi r^2 t l} = G r \phi$$

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Principle of virtual work.

- The principle of virtual work can be applied to torsion to give a compatibility relationship.
internal work = external work.

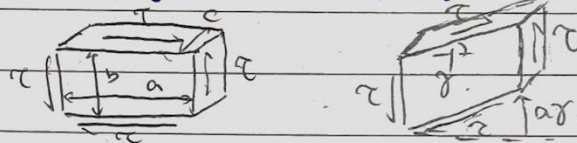
① External work.

- The external work is given by

$$\text{external work} = T\theta.$$

② Internal work.

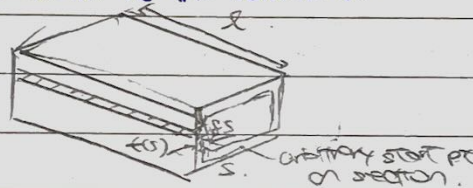
- Consider a small element, subject to a shear stress τ that undergoes a small shear strain γ .



The work done on the small element δW_i is given by

$$\begin{aligned}\delta W_i &= \tau \cdot (bc) \times (a\gamma) \\ &= \tau \gamma \delta V.\end{aligned}$$

Now consider a strip of the section as the volume δV



$$\delta V = l t(s) \delta s$$

→

$$\delta W_i = \tau(s) \cdot \gamma(s) l t(s) \delta s.$$

Integrating around the section,

$$W = \oint_C \tau(s) \gamma(s) l t(s) ds$$

since $q = \tau(s)t(s)$ is constant,

$$\text{internal work} = q l \oint_C \gamma(s) ds.$$

- Therefore, applying the principle of virtual work to torsion we have

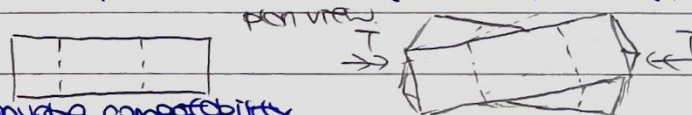
$$T\theta = q l \oint_C \gamma(s) ds$$

General thin-walled section

- Consider a general thin-walled section subject to an applied torque T .



- The compatibility between twist θ and shear strain γ is tricky because of warping. For non-circular cross-sections, plane-sections do not remain plane when twisted.



→ use virtual work to invoke compatibility.

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Equilibrium $\tau = \tau(s) t(s) = \frac{T}{2Ae} \rightarrow \tau(s) = \frac{q}{t(s)} = \frac{T}{2Ae t(s)}$

Compatibility $\theta = 2L \oint_{cs} \gamma(s) ds \rightarrow \phi = \frac{\theta}{L} = \frac{2 \oint_{cs} \gamma(s) ds}{L}$

Material law $\tau(s) = G \gamma(s)$

$$\therefore \frac{T}{2Ae t(s)} = G \gamma(s)$$

$$\gamma(s) = \frac{T}{2Ae G t(s)}$$

$$\rightarrow \phi = \frac{2Ae}{T} \oint_{cs} \frac{T}{2Ae G t(s)} ds$$

$$= \frac{2Ae}{G} \oint_{cs} \frac{1}{t(s)} ds$$

$$\therefore \frac{T}{\phi} = G \cdot \frac{4Ae^2}{\oint_{cs} t(s) ds}$$

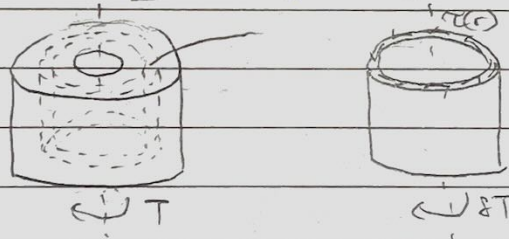
- The shear stress τ is given by

$$\tau = \frac{T}{2Ae t(s)} = \frac{1}{2Ae t(s)} G \frac{4Ae^2}{\oint_{cs} t(s) ds} \phi$$

$$\tau = G \left(\frac{2Ae}{t(s) \oint_{cs} t(s) ds} \right) \phi$$

Axisymmetric shafts

- For the special case of shafts w/ circular symmetry, the results for thin sections can be used to make calculations for thick, or solid sections.



For the thin shell,

$$\delta T = 2\pi r t \delta r G \phi$$

Integrating over the whole shaft,

$$T = 2\pi G \phi \int_{R_1}^{R_2} r t dr$$

$$= \frac{2\pi G \phi}{4} (R_2^2 - R_1^2)$$

$$\therefore \frac{T}{\phi} = G \left(\frac{\pi}{2} (R_2^2 - R_1^2) \right)$$

For circular shafts only, the torsion constant J is given by the polar second moment of area

$$J = \int r^2 dA$$

- For circular shafts, the shear stress τ is given by

$$\tau = G r \phi$$

and combining w/ $\frac{T}{\phi} = GJ$, we can say

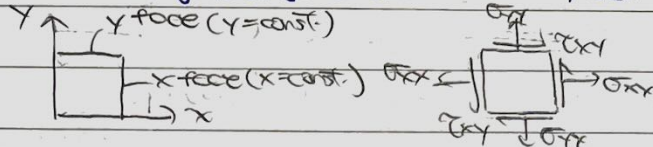
$$\frac{T}{r} = G \phi = \frac{T}{J}$$

cf for bending,
 $\frac{\sigma}{y} = E \kappa = \frac{M}{I}$

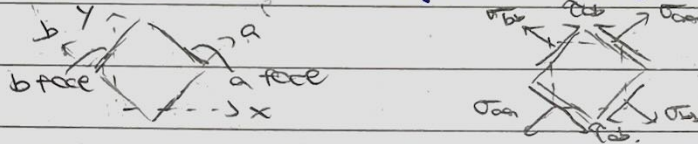
Analysis of stress

2D stress state.

- The following square shows a general state of stress in 2D, defined using axes x and y .



using another set of axes, a and b at an angle θ to the original set,



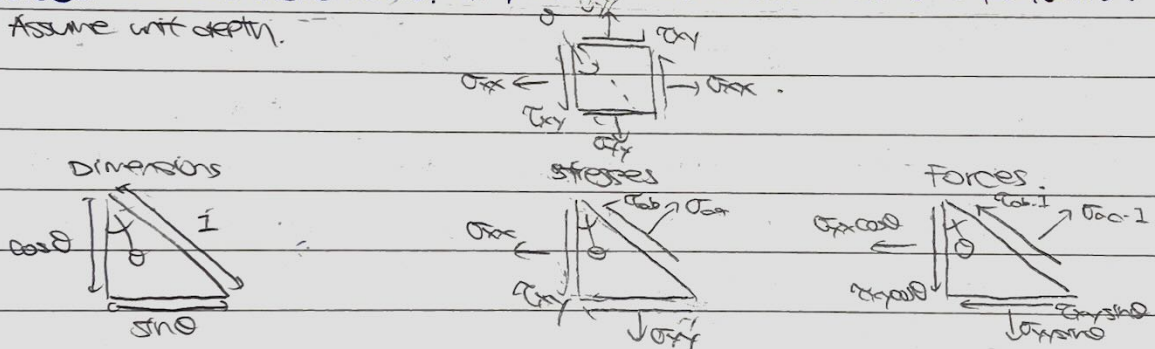
- Quick note on the notation for stresses.

$(\sigma/\tau)_{xy}$
 acting on x face $\rightarrow$
 acting in y direction

Equilibrium equations

- Consider the stress state below. Apply eqn. to the axis inclined at θ to the x -axis.

Assume unit depth.



$$N2L \nearrow \sigma_{aa} = (\sigma_{xx} \cos \theta + \tau_{xy} \sin \theta) \cos \theta + (\sigma_{yy} \sin \theta + \tau_{xy} \cos \theta) \sin \theta$$

$$\searrow \tau_{ab} = -(\sigma_{xx} \cos \theta + \tau_{xy} \sin \theta) \sin \theta + (\sigma_{yy} \sin \theta + \tau_{xy} \cos \theta) \cos \theta$$

Taking up, we obtain the equilibrium eqns.

$$\sigma_{aa} = \sigma_{xx} \cos^2 \theta + \sigma_{yy} \sin^2 \theta + 2\tau_{xy} \cos \theta \sin \theta$$

$$\tau_{ab} = -\sigma_{xx} \sin \theta \cos \theta + \sigma_{yy} \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)$$

* These eqns can be found in the structures PB.

Mohr's circle of stress in 2D.

- Plotting the stress state on a graph of shear stress against normal stress, we get Mohr's circle.

- For Mohr's circle the sign convention is that shear stress is +ve when it is acting clockwise.

- We can plot Mohr's circle using 2 pairs of normal and shear stress:

$$(\sigma_{xx}, \tau_{xy})$$

$$(\sigma_{yy}, \tau_{yx})$$

(A line connecting these two points is a diameter of Mohr's circle)

- starting w/ the equilibrium eqns,

$$\sigma_{aa} = \sigma_{xx} \cos^2 \theta + \sigma_{yy} \sin^2 \theta + 2\tau_{xy} \cos \theta \sin \theta.$$

$$\tau_{ab} = -\sigma_{xx} \sin \theta \cos \theta + \sigma_{yy} \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)$$

using $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$, $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$, $\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$, we get

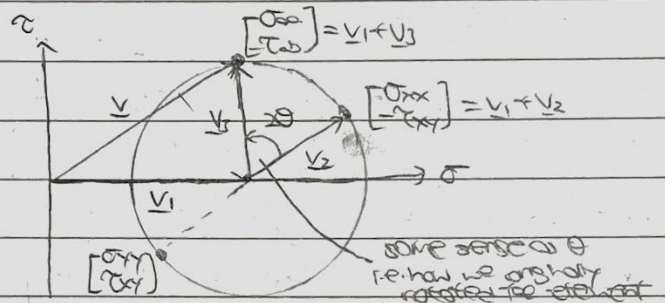
$$\sigma_{aa} = \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) + \frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{ab} = -\frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \sin 2\theta + \tau_{xy} \cos 2\theta.$$

this can be written as a vector eqn.

$$\begin{bmatrix} \sigma_{aa} \\ \tau_{ab} \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) \\ 0 \end{bmatrix}}_{V_1} + \underbrace{\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}}_{R(2\theta)} \underbrace{\begin{bmatrix} \frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \\ -\tau_{xy} \end{bmatrix}}_{V_2}$$

plotting the vectors,



- The stress state on two $\perp$ faces (e.g. x and y, a and b) lie at opposite ends of a diameter of the circle.
- The centre of the circle lies on the σ axis, at the mean value of the normal stresses on any $\perp$ faces.
- To find the stresses on faces at an orientation θ to the original faces, rotate around the Mohr's circle by 2θ in the same direction.
- There are two directions in which there is no shear stress on the $\perp$ faces - these are the principal directions of stress.

Principal stress directions,

- It is possible to show that for any state of stress in 3D, there will be three $\perp$ directions on which faces there are no shear stresses. - known as the principal stress directions.
- On these faces the only stress is a normal stress, and these three stresses are the principal stresses.
- + In the 2D case, the third principal stress direction is normal to the plane and the principal stress is zero.
- The principal stresses can be found easily using the Mohr's circle by considering the centre and the radius

$$C = \frac{\sigma_{xx} + \sigma_{yy}}{2}$$

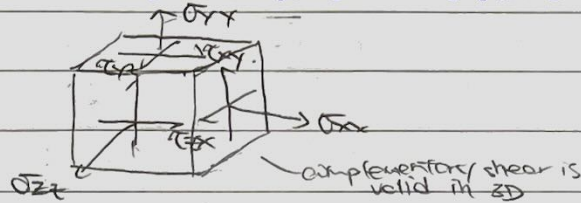
$$\rightarrow \sigma_1 = C + R$$

$$R = \sqrt{\left(\frac{\sigma_{xx} - \sigma_{yy}}{2}\right)^2 + \tau_{xy}^2}$$

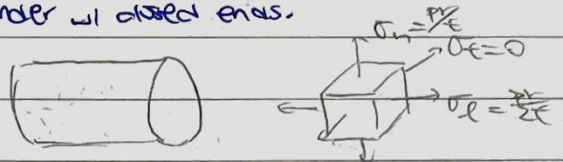
$$\sigma_2 = C - R.$$

3D stress state

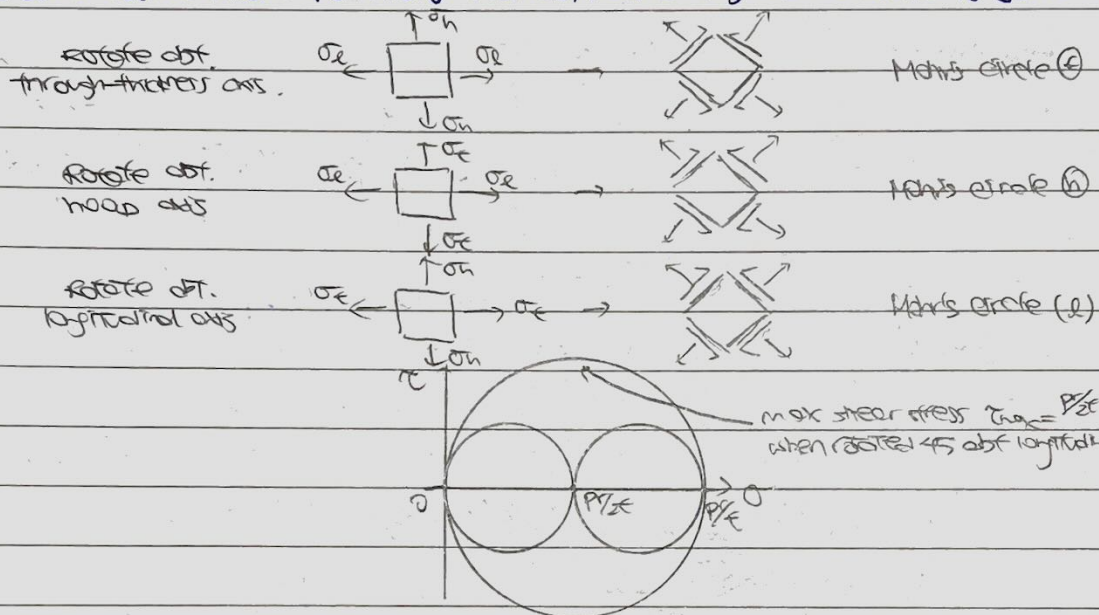
- In a 3D stress state, there will be three normal stresses and three shear stresses



- consider a pressurised cylinder w/ closed ends.



We could rotate about the through-thickness, hoop or longitudinal axis to get 3 Mohr's circles.



- We must only use Mohr's circle to rotate about principal axes — we would lose track of information about shear on the other faces.

(e.g. rotating about x-direction will calculate the change in $\sigma_{xx}, \sigma_{yy}, \tau_{yz}$, but lose track of change in τ_{xy}, τ_{zx} (this is fine if the x-direction is a principal stress direction because τ_{xy}, τ_{zx} would remain zero))

- If we know one of the principal stress states, we can rotate about that direction to find the other principal stresses → then complete the 3D Mohr's circle.

usually $\sigma_z = 0$.

stress tensor.

- the Mohr's circle only allows us to rotate around principal directions and doesn't give complete freedom in calculating stresses in any arbitrary direction. → use the stress tensor.

$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix}$$

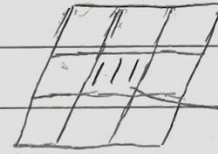
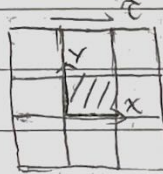
(details are beyond the scope of this course)

Analysis of strain

2D strain state

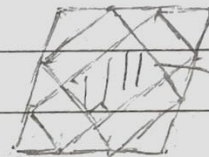
- As w/ stress, our measure of strain varies w/ the orientation of the axes used.

grid w/ lines
at $0^\circ, 90^\circ$



pure shear strain

grid w/ lines
at $45^\circ, 135^\circ$



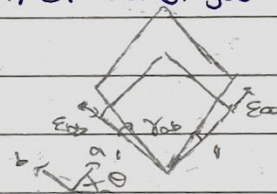
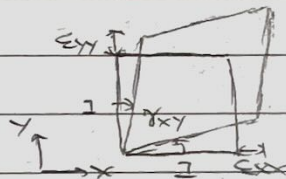
pure normal strain

- Quick note on the notation for strains

$(\epsilon/\gamma) \times y$ acting in y-direction
acting on x-face

Compatibility relationships

- The relationship b/w strain and axis orientation is simply a matter of geometry $\rightarrow$ compatibility.



It can be shown that

$$\epsilon_{x1x1} = \epsilon_{xx} \cos^2 \theta + \epsilon_{yy} \sin^2 \theta + \gamma_{xy} \cos \theta \sin \theta$$

$$\frac{\gamma_{x1y1}}{2} = -\epsilon_{xx} \sin \theta \cos \theta + \epsilon_{yy} \sin \theta \cos \theta + \frac{\gamma_{xy}}{2} (\cos^2 \theta - \sin^2 \theta)$$

$\rightarrow$ these eqns can be found in the structures DB.

- There is correspondence b/w the eqn relationships (stress) and the compatibility relationships (strain)

$$\epsilon_{xx} \leftrightarrow \sigma_{xx}; \quad \epsilon_{yy} \leftrightarrow \sigma_{yy}; \quad \frac{\gamma_{xy}}{2} \leftrightarrow \tau_{xy}; \quad \epsilon_{x1x1} \leftrightarrow \sigma_{x1x1}; \quad \epsilon_{y1y1} \leftrightarrow \sigma_{y1y1}; \quad \frac{\gamma_{x1y1}}{2} \leftrightarrow \tau_{x1y1}$$

- In the example where there is pure shear strain abt $0^\circ, 90^\circ$ axes, i.e.

$$\epsilon_{xx} = 0 \quad \epsilon_{yy} = 0 \quad \gamma_{xy} \neq 0$$

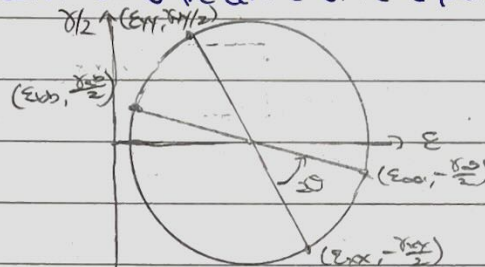
We see that abt the $45^\circ, 135^\circ$ axes, we have

$$\epsilon_{x45} = \frac{\gamma_{xy}}{2} > 0 \quad \epsilon_{y45} = -\frac{\gamma_{xy}}{2} < 0 \quad \gamma_{x45} = 0$$

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Mohr's circle of strain in 2D.

- Considering the correspondence b/w stress and strain formulae ($\epsilon \leftrightarrow \sigma$, $\frac{\gamma}{2} \leftrightarrow \tau$), we plot the strain state on a graph of $\frac{\text{shear strain}}{2}$ against normal strain, we get Mohr's circle.
- For Mohr's circle, the sign convention is that shear strain is +ve when the centre of the face has moved in a clockwise direction rel. to the centre of the square.



- The strain state in two $\perp$ directions (e.g. x and y, a and b) lie at opposite ends of a diameter of the circle.
- The centre of the circle lies on the ϵ axis, at the mean value of the normal strains of any $\perp$ directions.
- To find the strains in directions at an orientation θ to the original axes, rotate around the Mohr's circle by 2θ in the same direction.
- There are two directions in which there is no shear strain on the $\perp$ directions — these are the principal directions of strain.

Principal strain directions

- It is possible to show that for any state of strain in 3D, there will be three $\perp$ directions in which there is no shear strain — known as the principal strain directions.
- In these directions, the only strain is normal strain, and these three strains are the principal strains.
- The principal strains can be found easily using the Mohr's circle by considering the centre and the radius

$$C = \frac{\epsilon_{xx} + \epsilon_{yy}}{2}$$

$$R = \sqrt{\left(\frac{\epsilon_{xx} - \epsilon_{yy}}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

$$\rightarrow \epsilon_1 = C + R$$

$$\epsilon_2 = C - R$$

* For an isotropic material, no shear strain implies no shear stress, so the principal strain directions and the principal stress directions coincide.

- To find the third principal strain for plane stress ($\sigma_3 = 0$), we

→ 1) Find the first two principal strains ϵ_1, ϵ_2 (from $\epsilon_{xx}, \epsilon_{yy}, \gamma_{xy}$ using Mohr's circle).

→ 2) Find the first two principal stresses σ_1, σ_2

$$\epsilon_1 = \frac{1}{E} [\sigma_1 - \nu \sigma_2 - \nu \sigma_3^0]$$

$$\epsilon_2 = \frac{1}{E} [\sigma_2 - \nu \sigma_1 - \nu \sigma_3^0]$$

$$\sigma_1 = \frac{E}{1-\nu^2} (\epsilon_1 + \nu \epsilon_2)$$

$$\sigma_2 = \frac{E}{1-\nu^2} (\epsilon_2 + \nu \epsilon_1)$$

→ 3) Find the third principal strain ϵ_3

$$\epsilon_3 = \frac{1}{E} [\sigma_3^0 - \nu \sigma_2 - \nu \sigma_1]$$

zero σ_3 does not imply zero ϵ_3 !

3D strain state

- In a 3D strain state, there will be three normal strains and three shear strains.
- there is a 3D Mohr's circle of strain analogous to the 3D stress case, and also leads to loss of information unless the rotation of the axes is det. a principal does.
- (Refer to the section on 3D stress state for more details)

strain tensor.

- as for stress, there is a strain tensor to let us calculate strains in any arbitrary direction.

$$\begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{xy} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{xz} & \epsilon_{yz} & \epsilon_{zz} \end{bmatrix}$$

where ϵ_{xy} is the mathematical shear strain, defined as $\epsilon_{xy} = \frac{\gamma_{xy}}{2}$

strain gauges

- Resistance strain gauges rely on the fact that the resistance of a wire increases when stretched. To increase the sensitivity, the wire is run backwards and forwards, in a compact area.
- The change in resistance due to an applied strain is measured by a Wheatstone bridge.
- strain gauges can only measure normal strain along the length of the gauge and cannot measure shear strain directly.
- To measure the complete 2D state of strain at a pt, we req. three measurements from a strain gauge rosette.

① 45° rosette.

- The 45° rosette measures strains ϵ_0 , ϵ_{45} and ϵ_{90} . If the x-axis is aligned w/ the 0° direction,

$$\epsilon_{xx} = \epsilon_0 \quad ; \quad \epsilon_{yy} = \epsilon_{90}$$

Using the compatibility eqn $\epsilon_{\theta\theta} = \epsilon_{xx} \cos^2\theta + \epsilon_{yy} \sin^2\theta + \gamma_{xy} \cos\theta \sin\theta$, w/ $\theta = 45^\circ$, $\epsilon_{\theta\theta} = \epsilon_{45}$

$$\epsilon_{45} = \epsilon_0 \cdot \frac{1}{2} + \epsilon_{90} \cdot \frac{1}{2} + \gamma_{xy} \cdot \frac{1}{2} \quad \rightarrow \quad \gamma_{xy} = 2\epsilon_{45} - \epsilon_0 - \epsilon_{90}$$

② 60° rosette

- The 60° rosette measures strains ϵ_0 , ϵ_{60} , and ϵ_{120} . If the x-axis is aligned w/ the 0° direction,

$$\epsilon_{xx} = \epsilon_0$$

Using the compatibility eqn $\epsilon_{\theta\theta} = \epsilon_{xx} \cos^2\theta + \epsilon_{yy} \sin^2\theta + \gamma_{xy} \cos\theta \sin\theta$, w/ $\theta = 60^\circ, 120^\circ$, $\epsilon_{\theta\theta} = \epsilon_{60}, \epsilon_{120}$,

$$\epsilon_{60} = \epsilon_0 \cdot \frac{1}{4} + \epsilon_{yy} \cdot \frac{3}{4} + \gamma_{xy} \cdot \frac{\sqrt{3}}{4} \quad \rightarrow \quad \epsilon_{yy} = -\frac{1}{3}\epsilon_0 + \frac{2}{3}\epsilon_{60} + \frac{2}{3}\epsilon_{120}$$

$$\epsilon_{120} = \epsilon_0 \cdot \frac{1}{4} + \epsilon_{yy} \cdot \frac{3}{4} - \gamma_{xy} \cdot \frac{\sqrt{3}}{4} \quad \rightarrow \quad \gamma_{xy} = \frac{2\sqrt{3}}{3}\epsilon_{60} - \frac{2\sqrt{3}}{3}\epsilon_{120}$$

Von Mises yield criterion

- Von Mises yield criterion states that a body will yield when the strain energy of distortion reaches a critical value, ($U_0 = \frac{Y^2}{6G}$)

- The strain energy stored in a body can be split into hydrostatic and distortional strain energy. The distortional strain energy U_0 for a linear-elastic isotropic solid is given by

$$U_0 = \frac{1}{12G} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]$$

- In the uniaxial test case, ($\sigma_1 = Y, \sigma_2 = \sigma_3 = 0$), $U_0 = \frac{Y^2}{6G}$.

- The von Mises criterion can be written in terms of the principal stresses as follows:

At yield, $(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2Y^2$

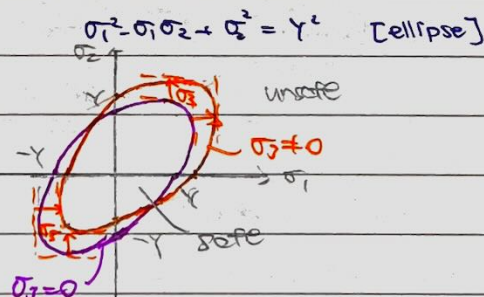
Visualising von Mises yield criteria.

- We can plot the yield surface on the interaction diagram to determine the sets of σ_1, σ_2 which will cause yield.

- For plane stresses, $\sigma_3 = 0$.

Uniaxial tests (1) $\sigma_1 = 0 \rightarrow \sigma_2 = \pm Y$; $\sigma_2 = 0 \rightarrow \sigma_1 = \pm Y$.

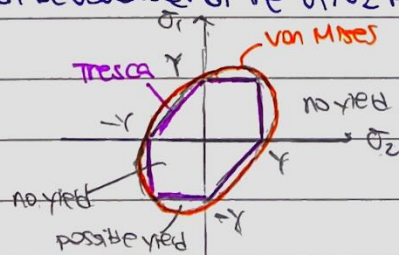
$$(2) (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2Y^2$$



- If $\sigma_3 \neq 0$, the basic shape of the yield criterion will be unaffected but the ellipse will be shifted to the right and up by σ_3 .

Comparison of Tresca and von Mises.

- If $\sigma_3 = 0$, the two criteria can be visualised on the σ_1, σ_2 plane as.



- The two criteria are in fairly good agreement, and the Tresca criterion is more conservative.

- The max discrepancy is for pure shear ($\sigma_1 = -\sigma_2$) where the difference $\sim 15\%$.

Elastic structural analysis

Principles of structural analysis

- There are three principles of structural mechanics that will be used repeatedly.
 - ↳ Equilibrium: Forces (stress, moments etc.) must always be in eqn.
 - ↳ Compatibility: Displacements (strains, curvatures etc.) must always be compatible.
 - ↳ Material laws: The constitutive laws (stress-strain, moment-curvature etc.) must follow a consistent material model (linear elastic (elasto-plastic)).
- For statically determinate problems, the forces can be solved by only considering eqn. The material law can then be used to find extensions and compatibility can be used to find displacements.
- For statically indeterminate problems, all three principles need to be applied in //.

Analysis of indeterminate truss structures

The force method for indeterminate truss structures:

① Indeterminacy: find the no. of redundancies n .

- Determine the no. of bars that needs to be removed for a statically determinate structure.
 - ↳ use structural intuition — one less than the no. of bars that needs to be removed for a mechanism

↳ use Maxwell's rule —
$$5M = b + r - J$$

Diagram labels:
 $5M$: mechanisms
 b : bars
 r : restraints
 J : joints
 $5M$ also labeled as redundancies (states of self stress)

② Equilibrium: find a general sol'n for the bar forces in eqn. w/ the applied load.

- Find a particular set of bar forces f_0 in eqn. w/ the applied loads. [particular sol'n]
 - ↳ Find the bar forces in eqn w/ applied loads when the redundant bar forces are set to zero.
- For each indeterminacy i , find a set of bar forces in eqn. w/ zero applied force [state of self stress]
 - ↳ Set the force to 1 in redundant bar i and 0 for other redundant bars, then find the other bar forces in eqn. w/ zero applied load.
- Write down the general eqn. sol'n, where x_i are unknowns to be solved.

$$f = f_0 + x_1 s_1 + x_2 s_2 + \dots + x_n s_n$$

Diagram labels:
 f : load
 f_0 : bar forces
 s_i : bar forces

(This is a sol'n to the matrix eqn. $f = \underline{C} x$, where $\underline{C}$ is the eqn. matrix)

③ Material: Find the extensions of each bar.

- The flexibility matrix $\underline{F}$ can be formed by writing the flexibility of each bar $\frac{L}{AE}$ along the diagonal.

- The extension e is the elastic extension $\underline{F}f$ plus any initial misfit e_0

$$e = \underline{F}f + e_0$$

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④ Compatibility: Find the ext. of each state of self-stress req. to ensure the structure fits together, i.e. x_i .

- Apply the principle of virtual work $\underline{\epsilon}^* \cdot \underline{e} = \underline{f}^* \cdot \underline{\delta}$ and use the virtual eqm. systems of states of self-stress ($\underline{\epsilon}^* = \underline{S}_i$, $\underline{f}^* = 0$), we get

$$\underline{S}_i \cdot \underline{e} = 0$$

(A diff eqn. can be found for each eqn. of self-stress $\rightarrow$ n simultaneous eqn. in n unknowns x_i).

$\rightarrow$ After solving for x_i , all the internal forces and extensions are known

$$\underline{\epsilon} = \underline{\epsilon}_0 + x_1 \underline{S}_1 + x_2 \underline{S}_2 + \dots + x_n \underline{S}_n$$

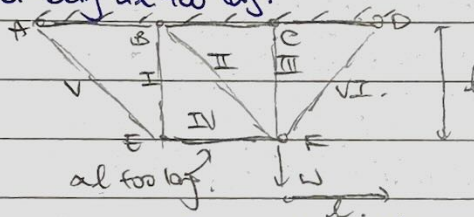
$$\underline{e} = \underline{F} \underline{\epsilon}_0 + x_1 \underline{F} \underline{S}_1 + x_2 \underline{F} \underline{S}_2 + \dots + x_n \underline{F} \underline{S}_n + \underline{e}_0$$

External displacements can be solved using virtual work $\underline{\epsilon}^* \cdot \underline{e} = \underline{f}^* \cdot \underline{\delta}$

given $\underline{f}^*$, find $\underline{\epsilon}^*$.
 $\underline{e}$ known, find $\underline{\delta}$
 (only if chosen over the same $\underline{f}^* \cdot \underline{e}$)

Example of the force method for indeterminate truss structure.

- Find the forces and extensions in all the members of the truss below due to loading and the initial mistake of bar EF being a bit too long.



① Indeterminacy: $S-M = b+r-2j = 6+8-2(6) = 2$

by inspection, $m=0$, so $s=2 \rightarrow$ choose V and VI as redundant.

② Equilibrium: set $t_v = t_{vi} = 0$ and solve for other t_i

$$\begin{array}{c} \text{Node E: } t_v=0, t_{vi}=0 \\ \text{Node F: } t_{vi}=0, t_{vi}=0 \end{array} \rightarrow \underline{\epsilon}_0 = \begin{bmatrix} 0 & 200 \\ 0 & 200 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \text{] redundant}$$

set $t_v=1$, $t_{vi}=0$, remove ext. loads, and solve for other t_i

$$\begin{array}{c} \text{Node E: } t_v=1, t_{vi}=0 \\ \text{Node F: } t_{vi}=0, t_{vi}=0 \end{array} \rightarrow \underline{\epsilon}_1 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

set $t_{vi}=1$, $t_v=0$, remove ext. loads, and solve for other t_i

$$\begin{array}{c} \text{Node E: } t_v=0, t_{vi}=1 \\ \text{Node F: } t_v=0, t_{vi}=1 \end{array} \rightarrow \underline{\epsilon}_2 = \begin{bmatrix} 0 \\ 1 \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

General sol'n of bar forces $\underline{f}$ in eqm. w/ applied loads $\underline{f}$ is

$$\underline{f} = \underline{f}_0 + x_1 \underline{S}_1 + x_2 \underline{S}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

no of dimensions

(Alternatively, we can find $\underline{f}_0, \underline{S}_1, \underline{S}_2$ by writing the eqm. matrix $\underline{C}$, $\underline{C} \underline{\epsilon} = \underline{f}$, $\underline{f}$ formed by d. eqn. of each node)

cannot identify redundant bars incorrectly, but more complicated.

$$\begin{bmatrix} 0 & 0 & 0 & -1/\sqrt{2} & 0 \\ -1 & 0 & 0 & 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 0 & 1 & 0 \\ 0 & -1/\sqrt{2} & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} f_{ex} \\ f_{ey} \\ f_{fx} \\ f_{fy} \end{bmatrix} = \begin{bmatrix} f_{ex} \\ f_{ey} \\ f_{fx} \\ f_{fy} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -W \end{bmatrix}$$

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③ Material: The extension e is a combination of elastic extensions $E\epsilon$ and initial misfit e_0 .

$$e = E(\epsilon_0 + x_1 s_1 + x_2 s_2) + e_0 = \frac{L}{AE} \begin{bmatrix} 1 & \sqrt{2} \\ 0 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 0 \\ 0.1 \\ 0.0 \\ 0.0 \end{bmatrix} + x_1 \begin{bmatrix} \sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

④ Compatibility: Apply virtual work on virtual eqn. systems of states of self stress.

$$s_1 \cdot e = 0 \rightarrow W(\frac{1}{\sqrt{2}})\frac{L}{AE} + x_1(2\sqrt{2} + \frac{3}{2})\frac{L}{AE} + x_2(-\sqrt{2}-1)\frac{L}{AE} + (\frac{1}{\sqrt{2}})L = 0$$

$$s_2 \cdot e = 0 \rightarrow -W(\sqrt{2})\frac{L}{AE} + x_1(-\sqrt{2}-1)\frac{L}{AE} + x_2(2\sqrt{2}+2)\frac{L}{AE} = 0$$

$$\therefore x_1 = 0.227 \frac{WL}{AE} ; x_2 = 0.293 \frac{WL}{AE} - 0.37 \frac{WL}{AE}$$

→ Force in the members $\underline{t}$

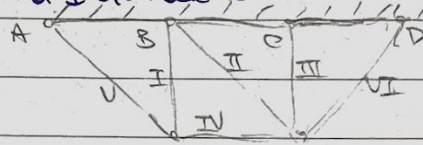
$$\underline{t} = \begin{bmatrix} 0 \\ 0.293 \\ 0.586 \\ 0 \\ 0 \\ 0.293 \end{bmatrix} W + \begin{bmatrix} 0.160 \\ 0.160 \\ 0 \\ -0.160 \\ -0.227 \\ 0.113 \end{bmatrix} \frac{WL}{AE}$$

Extension of the members $\underline{e}$

$$\underline{e} = \begin{bmatrix} 0.414 \\ 0.586 \\ 0 \\ 0 \\ 0.414 \end{bmatrix} \frac{WL}{AE} + \begin{bmatrix} 0.160 \\ 0.160 \\ 0 \\ -0.160 \\ -0.320 \\ 0.160 \end{bmatrix} \frac{WL}{AE}$$

- Find the vertical displacement of node E for the whole truss.

Apply a virtual vertical load of 1 at node E.



$$\underline{t}^* = \begin{bmatrix} t_{AB}^* \\ t_{BC}^* \\ t_{CD}^* \\ t_{DE}^* \\ t_{EF}^* \\ t_{FA}^* \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Solve for $\underline{t}^*$.

$$\underline{t}^* = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \underline{t}^* = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Apply virtual work to find δ_{eq} . $\underline{t}^* \cdot \underline{e} = \underline{t}^* \cdot \delta_{eq}$ (only $\underline{t}^*$ gives the same answer).

$$\underline{t}^* \cdot \underline{e} = 1(0 + 0.16 \frac{WL}{AE}) = 0.16 \frac{WL}{AE}$$

$$\underline{t}^* \cdot \underline{e} = -\sqrt{2}(0.414 \frac{WL}{AE} + 0.16 \frac{WL}{AE}) + 1(0.586 \frac{WL}{AE} + 0) + 1(0 + 0.8 \frac{WL}{AE}) + \sqrt{2}(0 - 0.32 \frac{WL}{AE}) = 0.16 \frac{WL}{AE}$$

$$\rightarrow \delta_{eq} = 0.16 \frac{WL}{AE}$$

Analysis of beams and frames

Three approaches to deflections in beams

① Differential eqn for bending ✓

- Equilibrium Find $M(x)$ for the given loading $w(x)$ either by

$$(a) S(x) = \int w(x) dx, M(x) = \int S(x) dx$$

(b). Free body analysis of an arbitrary cut x .

valid for long slender beams.

- Material Find $K(x)$ using the elastic law $K(x) = \frac{M(x)}{EI}$. (assume $S(x)$ cause no deformation)

- Compatibility Find the rotation $\theta(x)$ and deflection $v(x)$ using.

$$K(x) = \frac{d\theta}{dx} \approx \frac{d\theta}{dx} \rightarrow \theta(x) = \int K(x) dx$$

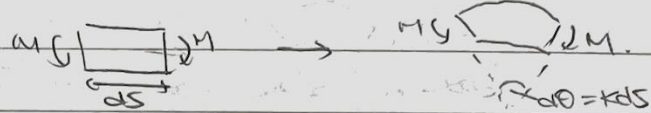
$$\theta(x) \approx -\frac{dv}{dx} \rightarrow v(x) = \int \theta(x) dx$$

② Virtual work ✓✓

- External work: for a beam loaded by pt-forces F_i , distributed force $w(s)$ and couples C_i , the work done by a compatible displacement δ / rotation θ_i will be

$$\text{ext. WD} = \sum F_i \delta_i + \int w(s) \delta(s) ds + \sum C_i \theta_i$$

- Internal work: Assuming beams undergo no axial/shear deformation $\rightarrow$ all work done by bending.



For a small element, the work done by the moment is $M d\theta = M \kappa ds$

For the complete beam, the work done will be

$$\text{int. WD} = \int M(s) \kappa(s) ds$$

- Equating the external and internal work, we get the eqn. of virtual work for a beam.

$$\sum F_i \delta_i + \int w(s) \delta(s) ds + \sum C_i \theta_i = \int M(s) \kappa(s) ds$$

$\hookrightarrow$ eqm. set: forces, couples and BMs must be in eqm.

$\hookrightarrow$ compatible set: displacements, rotations and curvatures must be compatible.

$\hookrightarrow$ the eqm / compatible sets do not have to be related, i.e. any compatible set can be imposed on any eqm. system. (we can have
 eqm: real ; compatible: virtual
 eqm: virtual ; compatible: real)

* Apply unit virtual force F^* to find displacements δ ; Apply unit virtual couple C^* to find rotations θ .

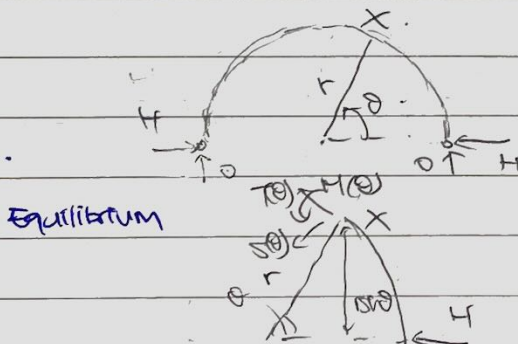
③ Database coefficients ✓✓✓

Example - deflection of a curved beam.
 curved $\rightarrow$ confuse DB cases
 $\rightarrow$ use method of virtual work.

- Find the deflection of B and rotation of A due to the applied load

Actual forces

Actual displacements



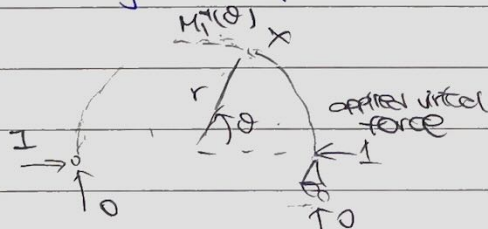
$$M_r: M(\theta) = H r \sin \theta$$

Material

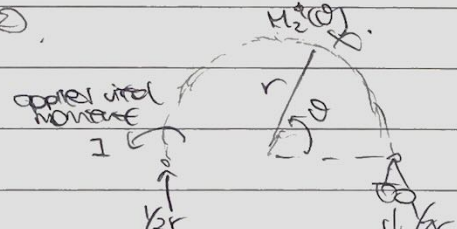
$$\kappa(\theta) = \frac{M(\theta)}{EI} = \frac{H r \sin \theta}{EI}$$

- consider the following virtual eqm. sets.

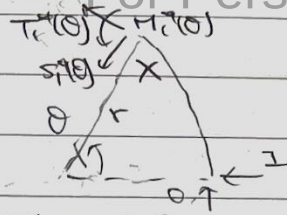
①



②

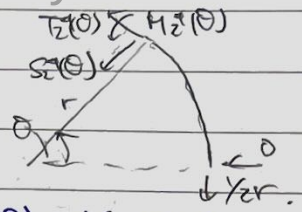


Equilibrium ①



$$M_1'(\theta) = r \sin \theta$$

②



$$M_2'(\theta) = \frac{1}{2}(1 - \cos \theta)$$

- Now find the req. displacement and reaction using virtual work

compatibility. ① $L \times h = \int M_1'(\theta) K(\theta) ds$

$$h = \int_0^{\pi} r \sin \theta \cdot \frac{H r \sin \theta}{EI} r d\theta = \frac{H r^3 \pi}{2EI}$$

② $L \times \phi = \int M_2'(\theta) K(\theta) ds$

$$\phi = \int_0^{\pi} \frac{1}{2}(1 - \cos \theta) \cdot \frac{H r \sin \theta}{EI} r d\theta = \frac{H r^2}{EI}$$

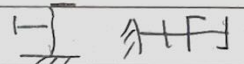
Identifying redundancies in beam and frame structures.

- Examples of determinate structures include:

→ cantilever



→ free structure



→ simply-supported beam

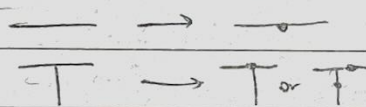


→ three-pinned arch



- Examples of removing redundancy include:

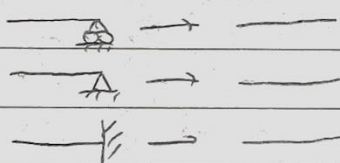
① Adding a pin.



destroy 1 moment → remove 1 redundancy

destroy 2 moments → remove 2 redundancies

② Removing a support.



destroy 1 force → remove 1 redundancy

destroy 2 forces → remove 2 redundancies

destroy 2 forces 1 moment → remove 3 redundancies

③ Cutting a beam



destroy $\frac{1}{2} BM$
 $\frac{1}{2} SF$
 $\frac{1}{2} Tension$ → remove 3 redundancies.

Force method for indeterminate beam and frame structures.

① Indeterminacy: Make the structure determinate (by adding pins/removing supports/cutting beam)

At such positions, explicitly add back the forces/moments that were destroyed.

→ statically determinate structure loaded by n unknown forces/moments.

② Equilibrium: Find all the internal forces in the structure in terms of the n unknowns

③ Material: Use the elastic law to find the internal deformation in the structure in terms of the n unknowns.

④ Compatibility: Find the displacements in terms of the n unknowns; then impose continuity

where pins were added / ensure displacements are zero where supports were

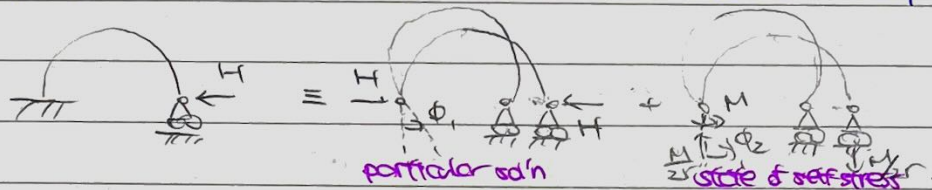
removed → give n eqns for the n unknown forces/moments.

+ Using deflection coefficients will combine steps ③ and ④.

↑ find int. forces (tension & BM)
deformations (curvature)
displacements / rotations.

Example: A ring structure

- Find the BM distribution in the frame due to the load H .

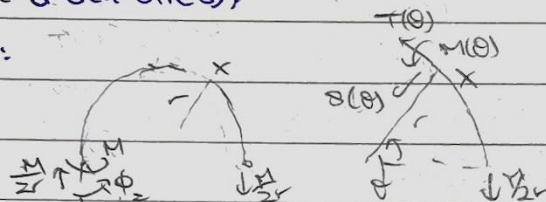


For the particular sol'n, as found previously,

$$\phi_1 = \frac{Hr^2}{EI}$$

For the state of self-stress,

Equilibrium:



$$M(\theta) = \frac{M}{2}(1 - \cos\theta)$$

Material:

$$K(\theta) = \frac{M(\theta)}{EI} = \frac{M(1 - \cos\theta)}{2EI}$$

Compatibility:

$$M\phi_2 = \int M(\theta)K(\theta)ds$$

$$M\phi_2 = \int_0^\pi \frac{M}{2}(1 - \cos\theta) \cdot \frac{M(1 - \cos\theta)}{2EI} r d\theta$$

$$\phi_2 = \frac{3\pi}{8} \frac{Mr}{EI}$$

The real system has no rotation at A, so

$$\phi_1 + \phi_2 = 0$$

$$\frac{Hr^2}{EI} + \frac{3\pi}{8} \frac{Mr}{EI} = 0$$

$$\therefore M = -\frac{8Hr}{3\pi}$$

The BM distribution is thus

$$M(\theta) = Hr \sin\theta + \frac{M}{2}(1 - \cos\theta)$$

$$= Hr \left(\sin\theta - \frac{4}{3\pi} + \frac{4}{3\pi} \cos\theta \right)$$

Example: A multi-span beam.

- Find the reaction forces at the supports of the multi-span beam



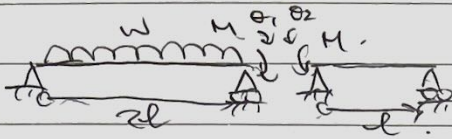
Add a pin (+ moment) at pin B to remove the redundancy



The above method works or used, but we can use the following trick for multi-span beams.



Apply DB cases.



$$\theta_1 = \frac{M(l)}{3EI} - \frac{w(2l)^2}{24EI}$$

$$\theta_2 = \frac{Ml}{3EI}$$

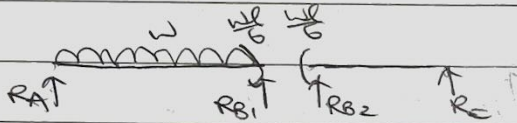
The red system is continuous over supports, so

$$\theta_1 + \theta_2 = 0$$

$$\frac{2Ml}{3EI} - \frac{wl^2}{6EI} + \frac{Ml}{3EI} = 0$$

$$M = \frac{wl^2}{6}$$

The reaction forces are thus:



$$R_A = \frac{3w}{2}, R_C = -\frac{w}{6}$$

$$R_{B1} = \frac{3w}{2}, R_{B2} = \frac{w}{6} \rightarrow R_B = R_{B1} + R_{B2} = \frac{5w}{2}$$

Implications on redundancy

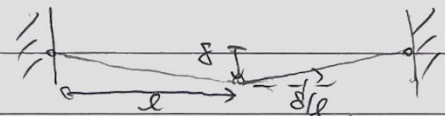
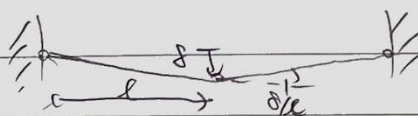
- For indeterminate structures, it is possible to have internal forces within the structure w/ no ext. loading applied (not free to move $\rightarrow$ state of self-stress). This may be due to
 - $\hookrightarrow$ settlement of supports
 - $\hookrightarrow$ temp. changes $\rightarrow$ thermal expansion/contraction.
 - $\hookrightarrow$ initial misfit during assembly
- For determinate structures, the structure could deform to take account of these effects.
- ∇ Such as are the same as the above but the particular joint is zero (unlocked anyway).

Application of self-stress

- Getting up a state of self-stress (prestress in either tension or compression) can be an important aspect of structural behaviour
- Application of self-stress include -
 - $\hookrightarrow$ Bicycle wheels: pre-tensioned spokes prevent buckling under loading.
 - $\hookrightarrow$ Concrete beams: concrete in compression due to pre-stressed steel tendons.
 - $\hookrightarrow$ Tensegrity systems provide stiffness due to prestress.

No prestress

w/ prestress T



$$F = 2T \sin(\theta/2) \approx 2T \theta/2$$

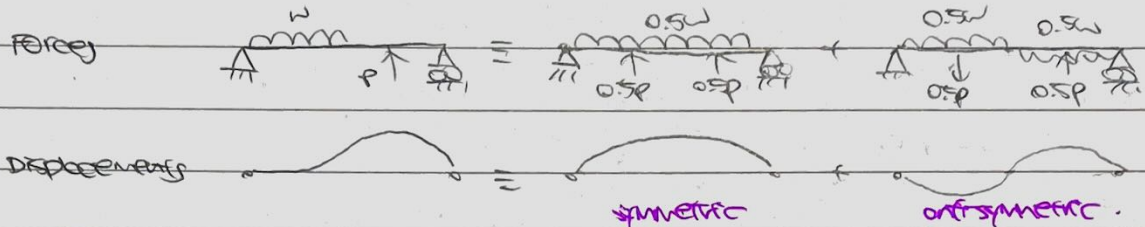
no resistance to motion $\rightarrow$ mechanism

$$\text{stiffness} = \frac{F}{\delta} = \frac{2T \theta/2}{\delta} = \frac{2T}{l}$$

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Analysis of symmetric structures.

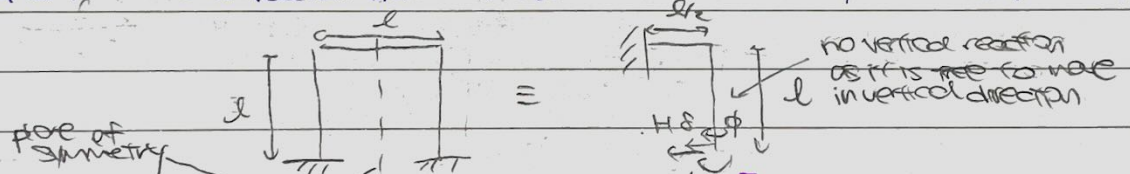
- We will consider structures w/ bilateral symmetry — it has a single plane of reflection.
- Anything symmetric is preserved by reflection of the structure in its plane of symmetry.
- Anything antisymmetric is reversed by reflection of the structure in its plane of symmetry.
- Any antisymmetric load can be split into a symmetric and antisymmetric component.



- A symmetric structure subject to (anti)symmetric loads will only have (anti)symmetric internal forces and (anti)symmetric displacements. This implies:
 - ↳ symmetric load → no shear, no rotation in plane of symmetry
 - ↳ antisymmetric load → no moment, no deflection in plane of symmetry

Example — self-stress of portal frame.

- In the plane portal frame, the flexural rigidity EI is uniform and coefficient of thermal expansion is α . Find the horizontal reaction and BM induced at the column feet by uniform temp. rise T .



using DB case:
$$\delta = \underbrace{\frac{(M+Hl)(\frac{l}{2})}{EI}}_{\phi_B} \cdot l + \underbrace{\frac{Hl^3}{3EI}}_{\delta_C \text{ due to } H} + \underbrace{\frac{Ml^2}{2EI}}_{\delta_D \text{ due to } M} - \alpha \frac{l}{2} T$$

$$\phi = \underbrace{\frac{(M+Hl)(\frac{l}{2})}{EI}}_{\phi_B} + \underbrace{\frac{Hl^2}{2EI}}_{\phi_C \text{ due to } H} + \underbrace{\frac{Ml}{EI}}_{\phi_D \text{ due to } M}$$

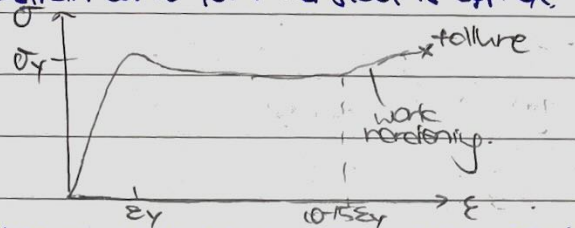
- solving self-stress as:
- 1) Remove restraints, add forces/moments.
 - 2) Solve for forces/moments using compatibility.

compatibility: $\delta = 0, \phi = 0 \rightarrow M = -\frac{2EI\alpha T}{l}, H = \frac{3EI\alpha T}{l^2}$

Plastic structural analysis

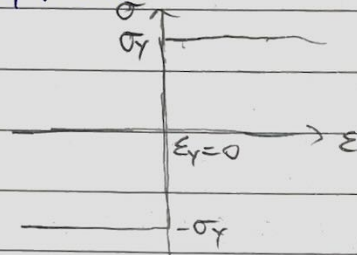
Material model

- In a tension test, the stress-strain curve for mild steel is approx.

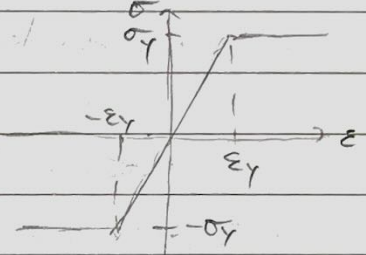


- In simple plasticity theory, this is idealised as either rigid perfectly plastic or elastic perfectly plastic.

rigid perfectly plastic



elastic perfectly plastic



- The main assumptions are:

- ↳ Neglect work-hardening (work fail)
- ↳ Material is ductile (no yield decrease)
- ↳ The behaviour is identical in tension and compression.

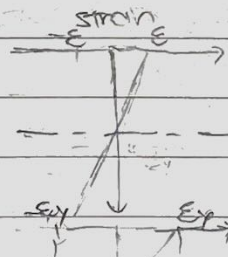
stresses on a rectangular cross-section as a plastic hinge forms.

- We make two assumptions about the behaviour of the beam:

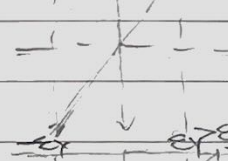
- ↳ Plane sections remain plane: $\epsilon = k y$.
- ↳ Material is elastic perfectly plastic: $\sigma = E \epsilon, \epsilon \leq \epsilon_y$; $\sigma = \sigma_y, \epsilon \geq \epsilon_y$

- For a simple rectangular section, the behaviour is as follows:

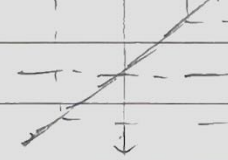
(a) Elastic: $\sigma = E \epsilon$



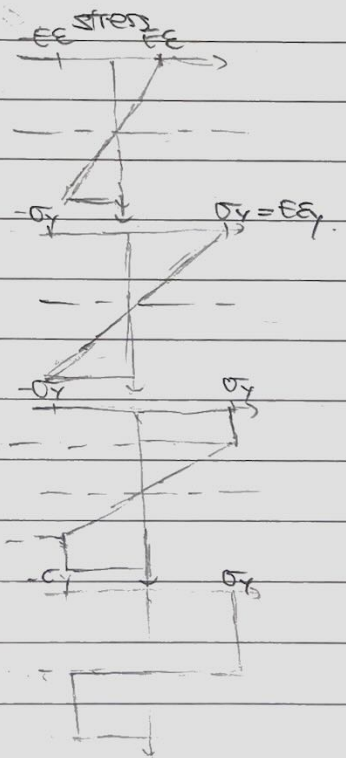
(b) First yield: $\sigma_y = E \epsilon_y$



(c) Partially plastic



(d) Fully plastic (limit)



(a) Linear elastic range ($M = EIK$).

- from elastic bending theory,

$$M = EIK, \quad (I = \frac{bd^3}{12} \text{ for rectangular section})$$

this is valid until first yield, when

$$|\sigma_{max}| < \left| \frac{M y_{max}}{I} \right| \leq \sigma_y, \quad (y_{max} = \frac{d}{2} \text{ for rectangular section})$$

(b) At first yield ($M = M_y$)

- At first yield,

$$|\sigma_{max}| = \left| \frac{M y_{max}}{I} \right| = \sigma_y \rightarrow M_y = \frac{I}{y_{max}} \sigma_y = z_e \sigma_y.$$

depends on cross-section depends on material

For a rectangular section,

$$z_e = \frac{bd^3/12}{d/2} = \frac{bd^2}{6} \rightarrow M_y = \frac{bd^2}{6} \sigma_y$$

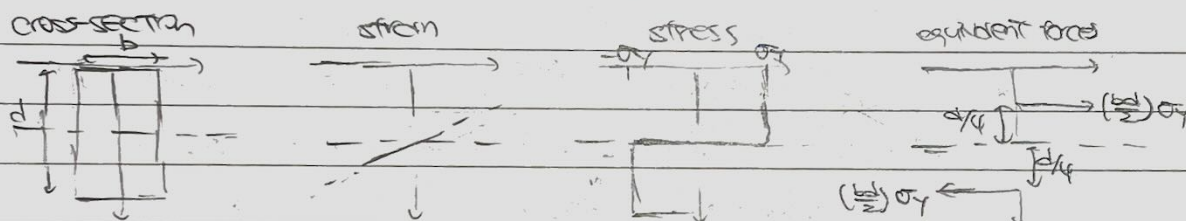
(c) Partially plastic ($M_y < M < M_p$)

- It can be shown for a rectangular section in the partially plastic range,

$$M = M_p \left[1 - \frac{1}{3} \left(\frac{k_y}{R} \right)^2 \right]$$

where k_y is the curvature at first yield.

(d) Fully plastic ($M = M_p$)



- the fully plastic moment M_p is in eqn. w/ the resultant forces.

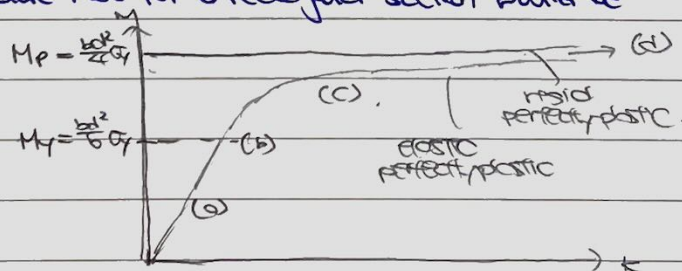
$$M_p = \frac{bd}{2} \sigma_y \frac{d}{4} + \frac{bd}{2} \sigma_y \frac{d}{4} = \frac{bd^2}{4} \sigma_y = z_p \sigma_y$$

depends on cross-section depends on material

for a rectangular section,

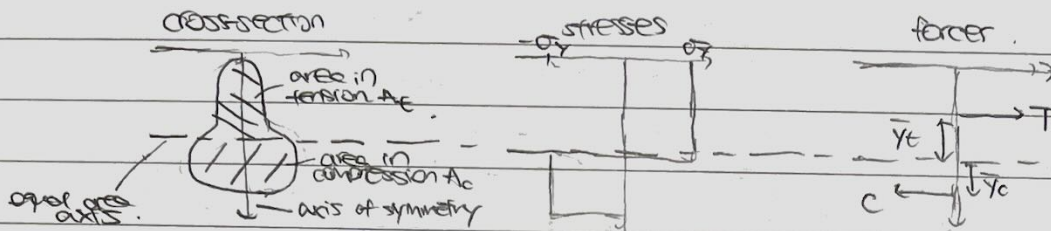
$$z_p = \frac{bd^2}{4} \rightarrow M_p = \frac{bd^2}{4} \sigma_y$$

→ the moment-curvature plot for a rectangular section would be



Plastic section modulus

- Consider a general cross section w/ an axis of symmetry in the dir. of the applied load.



The total tensile force T / compressive force C is given by

$$T = A_t \sigma_y$$

$$C = A_c \sigma_y$$

In the absence of any overall axial force (assumption), the forces must balance.

$$T = C$$

→

$$A_t = A_c$$

Thus the neutral axis for fully plastic bending is the equal-area axis.

- The plastic moment M_p is given by

$$M_p = \int_A y \sigma_y dA + \int_C (-y)(-\sigma_y) dA = \int_{\text{section}} |y| \sigma_y dA$$

hence the plastic section modulus Z_p is given by

$$Z_p = \int_{\text{section}} |y| dA$$

diff. from the first moment of area.

In practice, sections can be split into a no. of simple regions, so Z_p is given by

$$Z_p = \sum_{\text{regions}} A_r |\bar{y}_r|$$

where A_r is the area of a region yielded fully in tension / compression.

and $|\bar{y}_r|$ is the distance from the equal area axis to the centroid of A_r .

Plastic collapse of beams and frames (Upper bound analysis)

Upper bound theorem for plasticity.

- The upper bound thm for plasticity states that an estimate of the plastic collapse load W_{meth} calculated for any arbitrary compatible mechanism by equating the work done by the applied load, and the plastic energy dissipated, will be greater than or equal to the actual collapse load W_c .

$$W_{\text{meth}} \geq W_c$$

- The general methodology for finding the upper bound is as follows:

- 1) Postulate compatible collapse mechanisms (guess location of hinges)
- 2) Evaluate W_{meth} for each (using the material law)
- 3) select the collapse mechanism that gives the lowest upper bound on W_c .

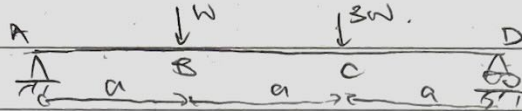
* Upper bound thm. uses compatibility and material law, but neglects eqn.

* All calculations neglect elastic deformation (assume rigid-perfectly plastic material)

Plastic collapse of statically determinate beams.

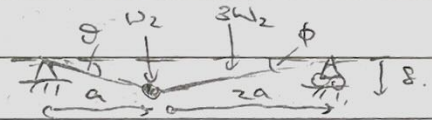
- A statically determinate beam can collapse when one hinge forms.

- Example 1: Find the collapse load of the simply supported beam below. The plastic moment of the beam is M_p .



① Mechanism 1 (Hinge at C)

Mechanism 2 (Hinge at B)



Compatibility of mechanism implies.

$$\theta = \frac{\delta}{2a}, \quad \phi = \frac{\delta}{a}$$

$$\theta = \frac{\delta}{a}, \quad \phi = \frac{\delta}{2a}$$

②. Work done by external loads. ($W_D = \text{force} \times \text{distance}$)

$$W_D = W_1 \theta a + 3W_1 \phi a$$

$$W_D = W_2 \theta a + 3W_2 \phi a$$

Energy dissipated by plastic hinge. ($E_D = \text{moment} \times \text{rotation}$)

$$E_D = M_p(\theta + \phi)$$

$$E_D = M_p(\theta + \phi)$$

Equating W_D by ext. load w/ E_D by plastic hinge, $W_D = E_D$.

$$W_1 \theta a + 3W_1 \phi a = M_p(\theta + \phi)$$

$$W_2 \theta a + 3W_2 \phi a = M_p(\theta + \phi)$$

$$W_1 a \left(\frac{\delta}{2a} + \frac{3\delta}{a} \right) = M_p \left(\frac{\delta}{2a} + \frac{\delta}{a} \right)$$

$$W_2 a \left(\frac{\delta}{a} + \frac{3\delta}{2a} \right) = M_p \left(\frac{\delta}{a} + \frac{\delta}{2a} \right)$$

$$W_1 = \frac{3M_p}{7a}$$

$$W_2 = \frac{3M_p}{5a}$$

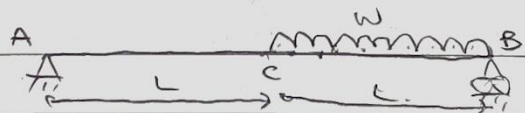
③..

$W_1 < W_2 \rightarrow W_1$ gives a lower upper bound

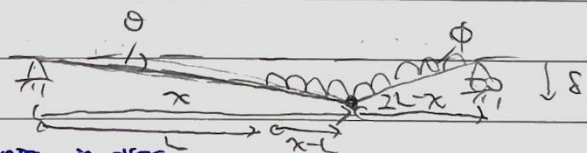
beam would collapse before the load can reach W_2 .

$$\therefore W_c \leq W_1 = \frac{3M_p}{7a}$$

- Example 2: Find the collapse load of the uniformly loaded simply supported beam. The plastic moment of the beam is M_p .



① consider a general collapse mechanism where a hinge forms at some distance x.



compatibility of mechanism implies.

$$\theta = \frac{\delta}{x}, \quad \phi = \frac{\delta}{2L-x}$$

② Work done by external load.

HB

$$W_{DHB} = W(2L-x) \cdot \frac{\phi(2L-x)}{2} = W(2L-x) \frac{\delta}{2}$$

CH

$$W_{DCH} = W(x-L) \theta \left(L + \frac{x-L}{2} \right) = W(x-L) \frac{L+x}{2} \frac{\delta}{x}$$

total

$$W_D = W_{DHB} + W_{DCH} = \frac{WL(2L-x)}{2x} \delta$$

Energy dissipated by plastic hinge

$$E_D = M_p(\theta + \phi)$$

Equating WD by ext. load w/ ED by plastic hinge, $W_D = E_D$,

$$\frac{WL(2x-L)}{2x} \delta = M_p(\theta + \phi)$$

$$\frac{WL(2x-L)}{2x} \delta = M_p \left(\frac{\delta}{x} + \frac{\delta}{2L-x} \right)$$

$$W = \frac{4M_p}{(2L-x)(2x-L)}$$

③. To find the min. possible W_{when} , consider when $\frac{dW_{when}}{dx} = 0$.

Equivalently, consider when $\frac{d(W_{when})}{dx} = 0$ for mathematical convenience.

$$\frac{1}{W_{when}} = \frac{1}{4M_p} [4Lx - 2x^2 - 2L^2 + Lx]$$

$$\frac{d}{dx} \left(\frac{1}{W_{when}} \right) = \frac{5L - 4x}{4M_p} = 0$$

$$\therefore x = \frac{5}{4}L$$

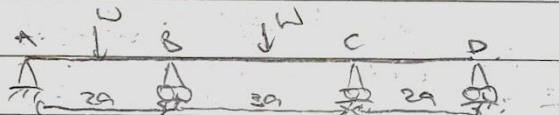
Substitute $x = \frac{5}{4}L$ into the expression for W_{when} to find W_c .

$$W_c = \frac{4M_p}{(3/2)(3/2)L} = \frac{32M_p}{7L^2}$$

Plastic collapse of statically indeterminate beams.

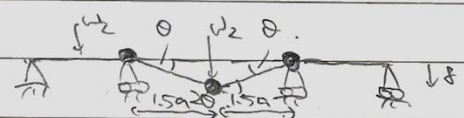
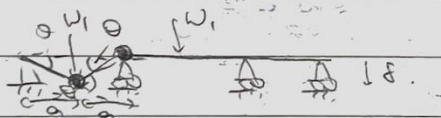
- For a statically indeterminate beam, failure must occur at more than one pt.

- Example 1: Find the collapse load of the multi-span beam below. The plastic moment of the beam is M_p .



① Mechanism 1 (Hinge at AB)

Mechanism 2 (Hinge at BC)



Compatibility of mechanism implies.

$$\theta = \frac{\phi}{2}$$

$$\theta = \frac{2\phi}{3a}$$

② Work done by external load.

$$W_D = W_1 \theta a$$

$$W_D = W_2 \theta \frac{3a}{2}$$

Energy dissipated by plastic hinge.

$$E_D = M_p(\theta + 2\theta)$$

$$E_D = M_p(\theta + \theta + 2\theta)$$

Equating WD by ext. load w/ ED by plastic hinge, $W_D = E_D$,

$$W_1 \theta a = M_p(\theta + 2\theta)$$

$$W_2 \theta \frac{3a}{2} = M_p(\theta + \theta + 2\theta)$$

$$W_1 = \frac{3M_p}{a}$$

$$W_2 = \frac{8M_p}{3a}$$

③.

$W_2 < W_1 \rightarrow W_2$ gives a lower upper bound

$$\therefore W_c \leq W_2 = \frac{8M_p}{3a}$$

Plastic collapse of plane frames

- This section considers the collapse of (pitched-roof) portal frames. The loading is often simplified as two pt. loads — a horizontal load H (wind load) and a vertical load V (snow load).

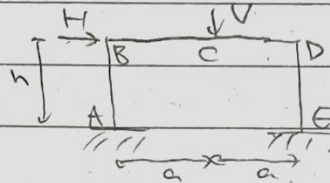
- The following calculations assume MCF.

↳ Buckling does not occur (checked for separately in practice).

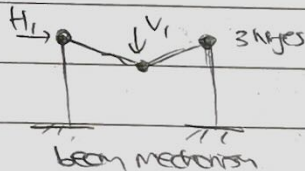
↳ Neglect effect of axial load on plastic moment M_p / plastic section modulus Z_p .

(a) Portal frame.

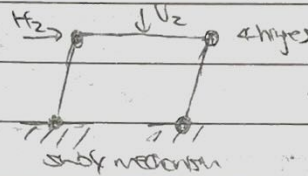
- Consider a portal frame subjected to the loads V and H as below:



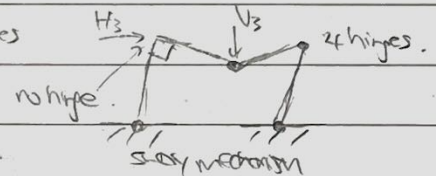
The feasible collapse mechanisms are as follows.



beam mechanism

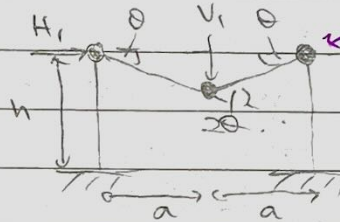


sway mechanism



sway mechanism

① Beam mechanism



for portal frames w/ nonuniform M_p , hinge forms in the weaker member (lower M_p), so ED is calculated using the smaller M_p for that hinge.

compatibility

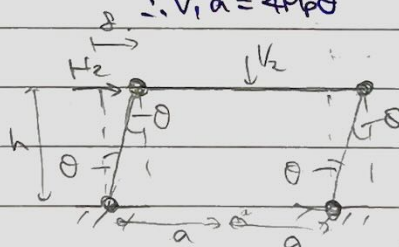
$$\delta = a\theta$$

material (WD=EP)

$$V_1 \delta = M_p(\theta + \theta + 2\theta)$$

$$\therefore V_1 a = 4M_p\theta$$

② Sway mechanism



compatibility

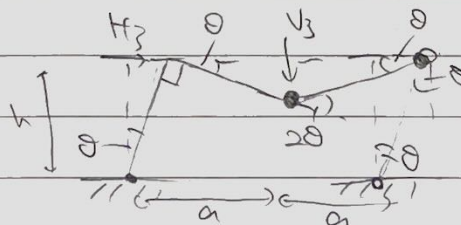
$$\delta = h\theta$$

material (WD=EP)

$$H_2 \delta = M_p(\theta + \theta + \theta + \theta)$$

$$\therefore H_2 h = 4M_p\theta$$

③ Combined mechanism



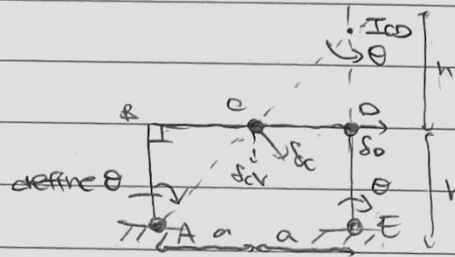
compatibility
material (WD=EP)

$$\delta_1 = a\theta \quad \delta_2 = h\theta$$

$$V_3 \delta_1 + H_3 \delta_2 = M_p(\theta + 2\theta + 2\theta + \theta) = 6M_p\theta$$

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The geometry for the combined mechanism is difficult $\rightarrow$ use method of ICS.

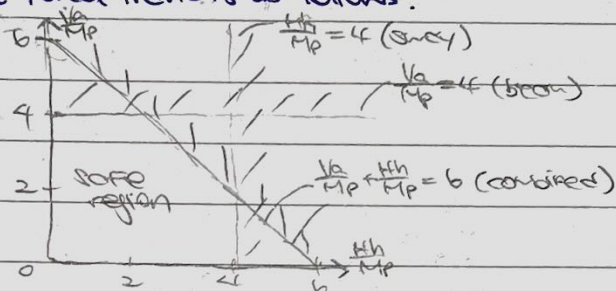


Define $\theta_{ABC} = \theta \rightarrow \theta_{AB} = \theta, \theta_{BC} = \theta$.

$\therefore \theta_A = \theta_{AB} = \theta, \theta_E = \theta_{DE} = \theta, \theta_C = \theta_{BC} + \theta_{CD} = 2\theta, \theta_D = \theta_{CD} + \theta_{DE} = 2\theta$

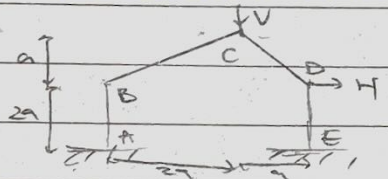
By geometry, $\delta_{cv} = a\theta, \delta_o = h\theta$ (for vertical δ_c , use horizontal lever distance $\times$ rotation)

- The interaction diagram of the portal frame is as follows.

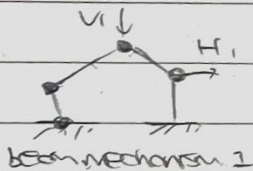


(b) PITCHED ROOF PORTAL FRAME.

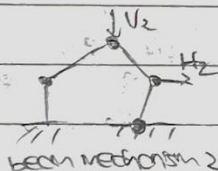
- consider a pitched roof portal frame subjected to the loads V and H as below:



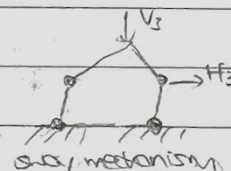
The feasible collapse mechanisms are as follows:



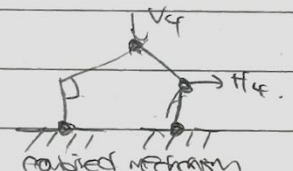
beam mechanism 1



beam mechanism 2



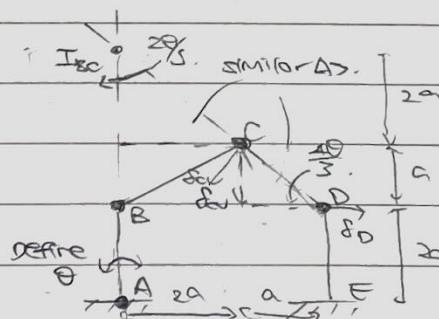
sway mechanism



combined mechanism

① Beam mechanism 1..

using the method of ICS,



Define $\theta_{AB} = \theta \rightarrow \theta_{BC} = \frac{2\theta}{3}, \theta_{CD} = \frac{4\theta}{3}$

$\therefore \theta_A = \theta_{AB} = \theta, \theta_D = \theta_{CD} = \frac{4\theta}{3}, \theta_B = \theta_{AB} + \theta_{BC} = \frac{5\theta}{3}, \theta_C = \theta_{BC} + \theta_{CD} = 2\theta$

By geometry, $\delta_{cv} = a \cdot \frac{4\theta}{3}, \delta_o = 0$

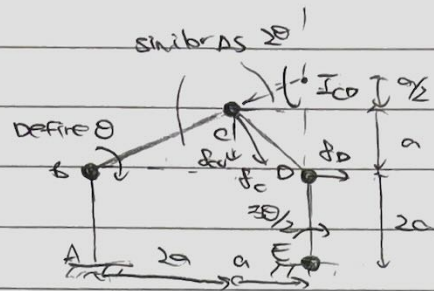
external (WD=ED)

$$V \delta_{cv} = M_p \left(\theta + \frac{4\theta}{3} + \frac{4\theta}{3} + 2\theta \right)$$

$$\therefore V \cdot a = \frac{9M_p}{2}$$

② Beam mechanism 2

Using the method of ICS,



Define $\theta_{BC} = \theta \rightarrow \theta_{CD} = 2\theta, \theta_{DE} = \frac{3\theta}{2}$

$\therefore \theta_B = \theta_{BC} = \theta, \theta_C = \theta_{DE} = \frac{3\theta}{2}, \theta_C = \theta_{BC} + \theta_{CD} = 3\theta, \theta_D = \theta_{CD} + \theta_{DE} = \frac{7\theta}{2}$

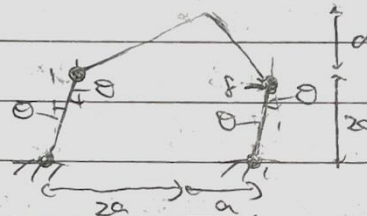
By geometry, $\delta_{CU} = 2a\theta, \delta_D = 2a \cdot \frac{3\theta}{2}$

material (WD=ED)

$$V_2 \delta_{CU} + H_2 \delta_D = M_p (\theta + \frac{3\theta}{2} + 3\theta + \frac{7\theta}{2})$$

$$\therefore 2V_2 a + 3H_2 a = 9M_p$$

③ sway mechanism



compatibility

$$\delta = 2a\theta$$

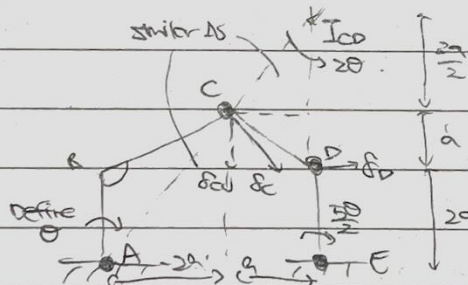
material (WD=ED)

$$H_3 \delta = M_p (\theta + \theta + \theta + \theta)$$

$$\therefore H_3 a = 2M_p$$

④ combined mechanism

Using the method of ICS,



Define $\theta_{BC} = \theta \rightarrow \theta_{CD} = 2\theta, \theta_{DE} = \frac{5\theta}{2}$

$\therefore \theta_A = \theta_{BC} = \theta, \theta_C = \theta_{DE} = \frac{5\theta}{2}, \theta_C = \theta_{BC} + \theta_{CD} = 3\theta, \theta_D = \theta_{CD} + \theta_{DE} = \frac{9\theta}{2}$

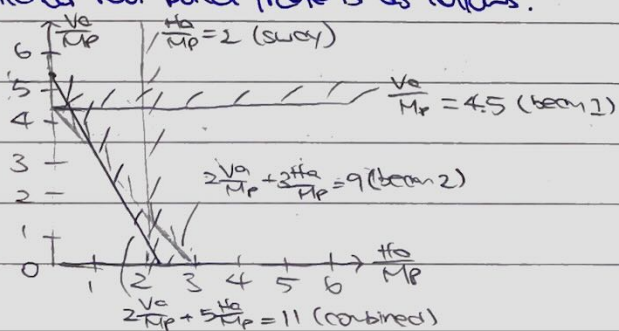
By geometry, $\delta_{CU} = 2a \cdot \theta, \delta_D = 2a \cdot \frac{5\theta}{2}$

material (WD=ED)

$$V_4 \delta_{CU} + H_4 \delta_D = M_p (\theta + \frac{5\theta}{2} + 3\theta + \frac{9\theta}{2})$$

$$2V_4 a + 5H_4 a = 11M_p$$

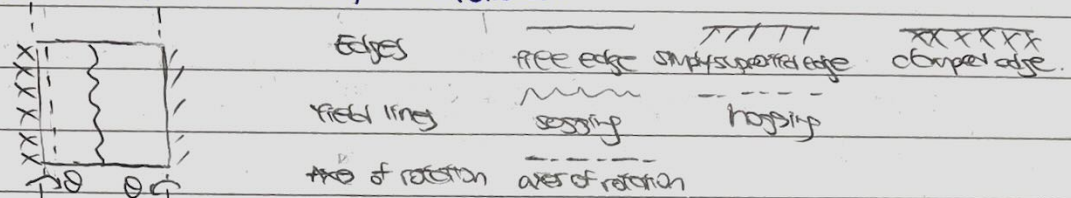
The interaction diagram of the pitched roof portal frame is as follows.



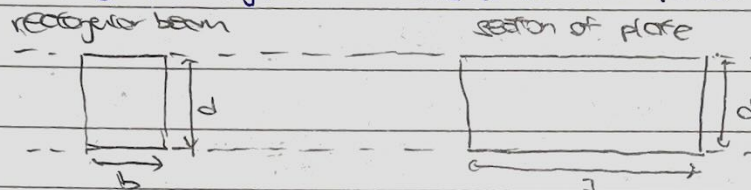
Yield line analysis of slabs and plates (upper bound analysis)

Yield line theory

- The upper-bound theorem can be extended from 1D beams and frames to 2D plate structures using yield line theory.
- The notation used for yield line theory is as follows:



- Consider the cross section of a rectangular beam and a section of plate.



As defined earlier, the plastic moment M_p in a rectangular beam is

$$M_p = Z_p \sigma_y = \frac{bd^2}{4} \sigma_y$$

For a plate, we define the moment capacity per unit length m as

$$m = \frac{d^2}{4} \sigma_y$$

In this course, we assume the value of m is isotropic (same in sagging/hogging) unless stated otherwise.

- Assumptions for yield line theory are as follows:

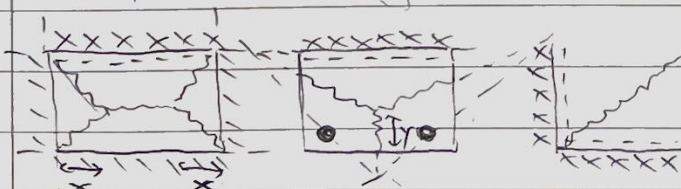
- ↳ collapse is due to brittle flexural failure (not other modes like shear failure)
- ↳ in-plane forces are ignored
- ↳ the deformations are small compared to the overall dimensions of the structure.

Compatible yield line patterns

- The following guidelines ensure that a chosen yield line pattern is compatible.

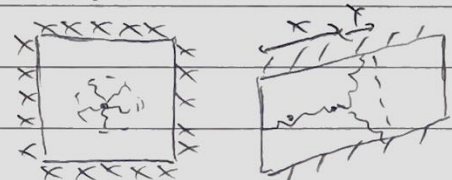
- ↳ The yield lines divide the slab into several rigid regions, which must remain planar. Each region must have a unique axis about which it rotates.
- ↳ The yield lines must be straight between intersections.
- ↳ A yield line between two rigid regions must pass through the intersection of the axis of rotation of those regions.

uniformly distributed loading



x, y are variable parameters.

pt. loads



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Work equation

- By assuming a compatible collapse mechanism, an upper bound for the collapse load of a slab can be calculated by equating the work done and energy dissipated during collapse.

- The work done by a pt. load $\downarrow$ displacing a distance δ is still given by

$$WD = W \cdot \delta$$

- The work done by a uniform pressure p resulting in a displacement $\delta(x,y)$ is given by

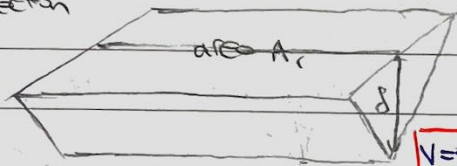
$$WD = \int p \delta(x,y) dA$$

$$= p \int \delta(x,y) dA$$

$$WD = pV$$

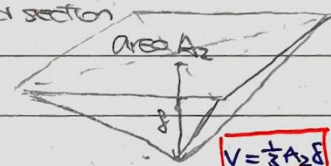
where V is the volume swept out by the collapsing slab. This can be easily calculated as most collapse mechanisms can be split into roof sections and pyramidal sections.

roof section



$$V = \frac{1}{2} A_1 \delta$$

pyramidal section



$$V = \frac{1}{3} A_2 \delta$$

- For a hinge b/wn two regions i and j , the total plastic moment in the hinge will be

$$M_p = m l_{ij}$$

where l_{ij} is the length of the hinge b/wn rigid regions i and j

- The energy dissipated in this hinge will therefore be

$$ED_{ij} = m l_{ij} \theta_{ij}$$

where θ_{ij} is the rotation of the hinge b/wn rigid regions i and j .

- The total energy dissipated will therefore be

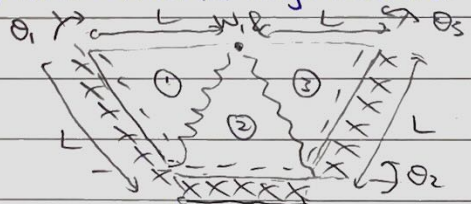
$$ED = \sum m l_{ij} \theta_{ij}$$

calculate for all hinges (edges which are hinged or jagged).

- Since we assume that all rotations are small, the rotations of each rigid region can be plotted as vectors, and the diff. b/wn these vectors is the rotation along the hinges.

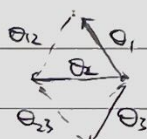
(All the rotation vectors pt. from the origin; only the magnitude of the diff. b/wn vectors is used in the calculation)

- Example: consider the balcony below. find the hinge rotations.



By geometry, $\delta = \frac{\sqrt{3}}{2} L \cdot \theta_i \rightarrow \theta_i = \frac{2\delta}{\sqrt{3}L}$, where $i=1,2,3$.

vector diagram of rotation

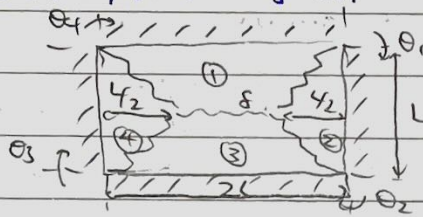


$$|\theta_{12}| = |\theta_{23}| = \frac{2\delta}{\sqrt{3}L}$$

θ_{12}/θ_{23} must be drawn st. // hinge 12 / hinge 23.

Example - simply-supported slab.

- Estimate the uniform pressure p req. to cause the collapse of the simply supported slab shown below, which has a moment capacity per unit length m .

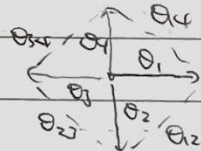


$$L_{13} = L$$

$$\frac{L_{ij}}{i \neq j} = \frac{1}{\sqrt{2}}$$

$$\delta = \frac{L}{2} \theta_i \rightarrow \theta_i = \frac{2\delta}{L}$$

vector diagram of rotation



$$\theta_3 = 2\theta_1 = \frac{4\pi}{L}$$

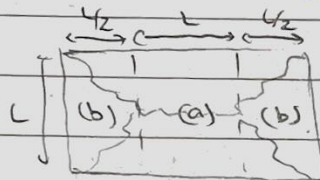
$$\theta_{ij} = \sqrt{2} \theta_i = \frac{2\sqrt{2}f}{L}$$

Energy dissipated.

$$E_D = m L_{13} \theta_{13} + \sum_{j \neq 13} m l_{ij} \theta_{ij}$$

$$= m L \cdot \frac{40}{L} + m \cdot \frac{L}{\sqrt{2}} \cdot \frac{2\sqrt{2}}{L} = 12m8$$

Work done.



$$\text{Volume (V)} = \frac{1}{2} L^2 \delta$$

$$\text{volume (b)} = 2 \times \frac{1}{3} \frac{1}{2} l^2 \delta = \frac{1}{3} l^2 \delta$$

$$\text{total volume} = \frac{1}{2}L^2\delta + \frac{1}{3}L^2\delta = \frac{5}{6}L^2\delta$$

$$WD = \rho V = \frac{5}{6} \rho L^2 \delta.$$

$$WD = ED:$$

$$\frac{57}{6} \text{ PL}^2 \delta = 12 \text{ m} \delta$$

$$P_{\text{mech}} = \frac{72 \text{ m}}{5 \text{ L}^2}$$

$$\therefore p \leq p_{\text{mean}} = \frac{F_{2m}}{5L^2}$$

p is found to be $p = 14.1 \text{ m}^2/\text{s}^2$.

Strip plane analysis of continua (Upper bound analysis)

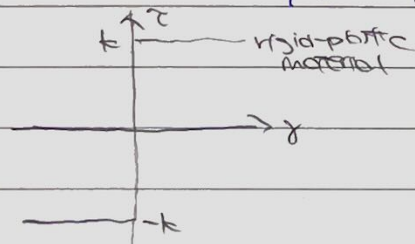
slip plane analysis.

- The upper bound theorem can be applied to the failure of rigid-plastic blocks of material (continuous) using slip plane analysis.

- Assumptions for slip plate analysis are as follows:

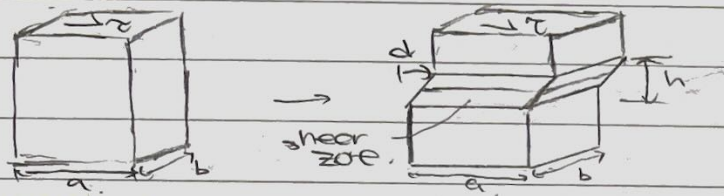
↳ rigid plastic material — the material will only fail in shear at a shearing stress k , and is otherwise rigid (example of Tresca yield criterion)

→ small deformations — geometry remains essentially unchanged.



Energy dissipated in shearing.

- The energy dissipated in slip plane mechanisms is dissipated in narrow bands of intense shearing.

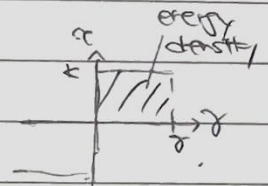


The shear strain γ of the shear zone is given by

$$\gamma = \frac{d}{h}$$

The energy dissipated per unit volume of shear zone is given by

$$ED/vol = k\gamma = k\frac{d}{h}$$



therefore the total energy dissipated is given by

$$ED = ED/vol \times V$$

$$= k\frac{d}{h} \cdot abh$$

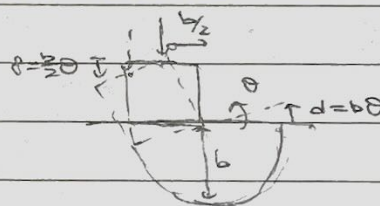
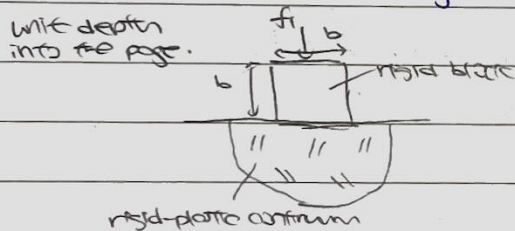
$$ED = ab \cdot k \cdot d$$

assume k is infinitesimally min.

* Note that the energy dissipated does not depend on the thickness of the shear zone h .

Slip-plane mechanism.

- consider a rigid block pressing onto a rigid-plastic continuum as follows.



work done

$$WD = f_1 \cdot \delta = \frac{f_1 b \theta}{2}$$

energy dissipated.

$$ED = \text{area} \cdot k \cdot d$$

$$= \pi b \cdot k \cdot b \theta = \pi b^2 k \theta$$

$$WD = ED:$$

$$\frac{f_1 b \theta}{2} = \pi b^2 k \theta$$

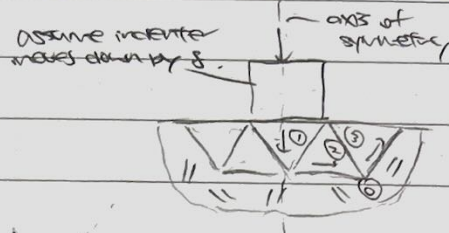
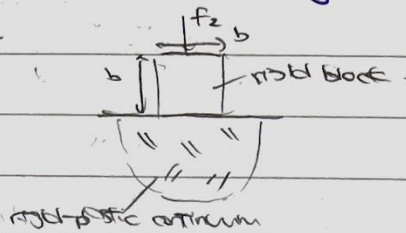
$$f_1 = 2\pi b k$$

$$\therefore f \leq f_1 = 2\pi b k$$

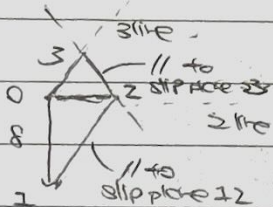
Triangular block mechanism

- consider a rigid block pressing onto a rigid-plastic continuum as follows.

unit depth
into the page



Displacement diagram.



$$d_{12} = \delta \cdot \frac{2}{\sqrt{3}}$$

$$d_{02} = d_{03} = d_{01} = \delta \cdot \frac{1}{\sqrt{3}}$$

work done.

$$WD = f_2 \cdot \delta$$

energy dissipated

$$ED = \sum \text{area} \cdot k \cdot d$$

$$= (l_{12} \cdot k \cdot d_{12} + l_{02} \cdot k \cdot d_{02} + l_{03} \cdot k \cdot d_{03} + l_{23} \cdot k \cdot d_{23}) \times 2$$

$$= \left(b k \frac{2\delta}{\sqrt{3}} + b k \frac{\delta}{\sqrt{3}} + b k \frac{\delta}{\sqrt{3}} + b k \frac{\delta}{\sqrt{3}} \right) \times 2 = \frac{10 b k \delta}{\sqrt{3}}$$

$$WD = ED,$$

$$f_2 \cdot \delta = \frac{10}{\sqrt{3}} b k \delta$$

$$f_2 = \frac{10}{\sqrt{3}} b k$$

$$\therefore f \leq f_2 = \frac{10}{\sqrt{3}} b k.$$

Rigid block - rigid-plastic continuum interface.

- some mechanisms involve sliding between the rigid block and the rigid-plastic continuum — and we don't know the properties of this interface → consider two limiting situations.

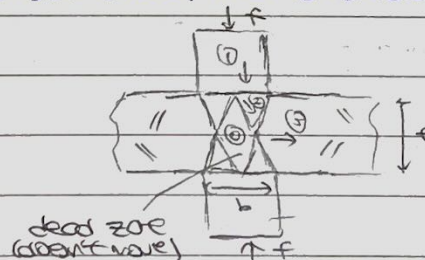
→ 1) zero friction — no energy is dissipated in the interface.

→ 2) Infinite friction — material next to the punch will shear as normal slip plane, and

$ED = \text{area} \cdot k \cdot d$, where k is the limiting shear stress of the rigid-plastic continuum.

- Example: consider the mechanism below, what is the total energy dissipated.

unit depth
into the page



zero friction

$$ED = \sum \text{area} \cdot k \cdot d$$

$$= (l_{02} \cdot k \cdot d_{02} + l_{23} \cdot k \cdot d_{23}) \times 4$$

infinite friction

$$ED = \sum \text{area} \cdot k \cdot d$$

$$= (l_{02} \cdot k \cdot d_{02} + l_{23} \cdot k \cdot d_{23} + \underbrace{l_{12} \cdot k \cdot d_{12}}_{\text{extra term (interface 12)}}) \times 4$$

extra term
(interface 12)

Plastic collapse of beams (Lower bound analysis)

Lower bound theorem for plasticity

elastic soln is an eqm. soln $\rightarrow$ eqm. soln that doesn't violate yield is safe, although it may not be an optimum design.

- The lower bound thm. for plasticity states that if a set of int. stresses can be found in the structure that are in eqm. w/ an applied load W_{eqm} , and nowhere violate the yield condition, then the applied load will be less than or equal to the actual collapse load W_c .

$$W_{eqm} \leq W_c$$

MUST have particular eqm. soln but can add as much as possible state of self stress to still require possible load W_c . $1M \leq M_p$ everywhere.

- The general methodology for finding the lower bound is as follows:
 - $\hookrightarrow$ 1) decompose the indeterminate beam into a determinate beam + unknown moments.
 - $\hookrightarrow$ 2) Find the BM diagram for the particular soln and state of self stress $\rightarrow$ overall BM diagram
 - $\hookrightarrow$ 3) choose unknown M_i s.t. $|M_{max}| = |M_{min}| = M_p \rightarrow$ calculate W_{eqm} / Z_p using material law.

(When designing beams, start from the weaker span first)

* Lower bound thm. uses eqm. and material law, but neglects compatibility.

* Using the lower bound thm. avoids the need to consider the detailed sequence of events leading to collapse $\rightarrow$ kinks will form as necessary at plastic hinges

Useful equilibrium results.

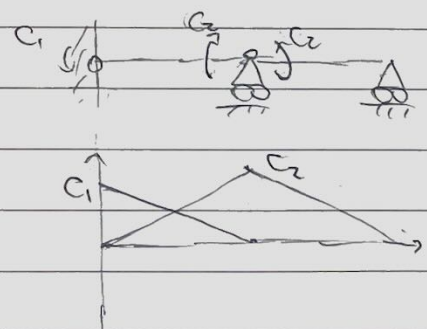
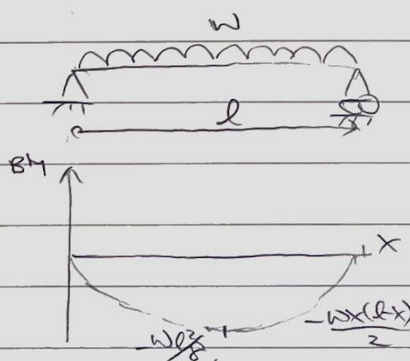
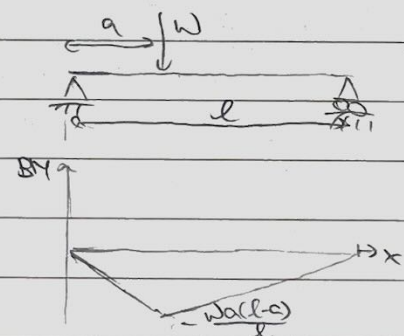
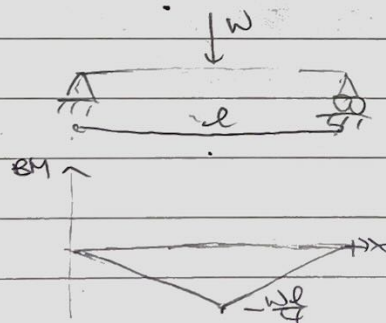
- For a simply-supported beam of length l :

$\hookrightarrow$ central pt. load W : $M_{max} = -\frac{Wl}{4}$ at the centre, M is piecewise linear

$\hookrightarrow$ pt. load W at a from support: $M_{max} = -\frac{Wa(l-a)}{l}$ at pt. load, M is piecewise linear.

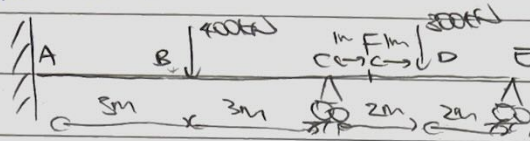
$\hookrightarrow$ distributed load w : $M_{max} = -\frac{wl^2}{8}$ at centre, $M(x) = -\frac{wx(l-x)}{2}$

$\hookrightarrow$ couple C at support ($x=0$): $M_{max} = C$ at support, M is (piecewise) linear.

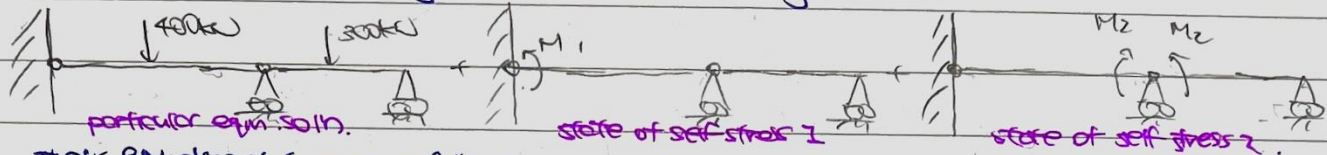


Example — two span beam.

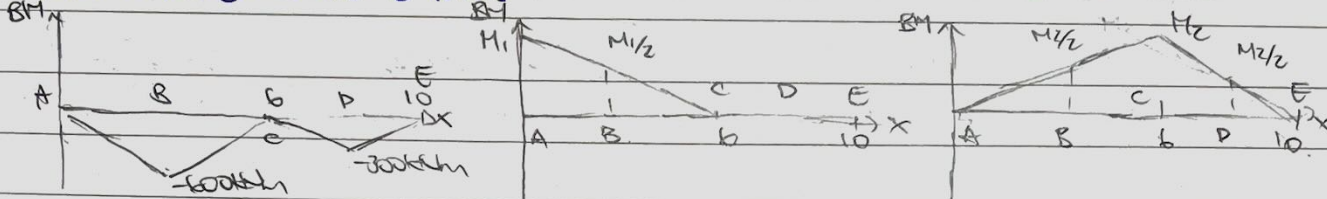
— Consider the following two-span beam.



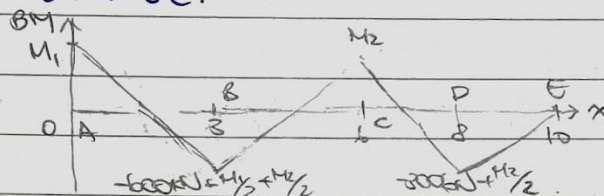
This is equivalent to adding pins and their corresponding moments.



Their BM diagrams are as follows



The overall BM diagram is therefore.



— Choose a UB suitable for the entire span AB [Design 1]

Max M at A and C, $M_A = M_C$, $M_C = M_2 \rightarrow$ optimum soln has $M_1 = M_2$.

Min. M at B and D, $M_B = -600 + \frac{M_1}{2} + \frac{M_2}{2}$, $M_D = -300 + \frac{M_2}{2} \rightarrow$ try B limiting, then check D.

Assume B limiting, set $M_{max} = |M_{min}|$, $-M_1 = -600 + M_1 \rightarrow M_1 = 300 \text{ kNm} (=M_2)$.

Now check D $M_D = -300 + \frac{M_2}{2} = -150 \text{ kNm} \rightarrow$ not limiting $\checkmark$.

$\therefore$ we ensure that $|M| \leq 300 \text{ kNm}$ everywhere in the entire structure. so choose Z_p of.

$$Z_p = \frac{M_p}{\sigma_y} \geq \frac{300 \text{ k}}{300 \text{ MPa}} = 1000 \times 10^6 \text{ m}^3 = 1000 \text{ cm}^3.$$

A suitable section would be UB-356x171x57, $Z_p = 1010 \text{ cm}^3$, $M_p = 303 \text{ kNm}$

— choose two UB suitable for the spans AC and CE, [Design 2]

Design the weaker beam CE first, set $M_{max} = |M_{min}|$, $-M_2 = -300 + \frac{M_2}{2} \rightarrow M_2 = 200 \text{ kNm}$.

Then design AC, set $M_{max} = |M_{min}|$, $-M_1 = -600 + \frac{M_1}{2} + \frac{200}{2} \rightarrow M_1 = 333 \text{ kNm}$.

$\therefore$ we ensure that $|M| \leq 333 \text{ kNm}$ in AC, so choose Z_p of.

$$Z_p = \frac{M_p}{\sigma_y} \geq \frac{333 \text{ k}}{300 \text{ MPa}} = 1111 \times 10^6 \text{ m}^3 = 1111 \text{ cm}^3$$

A suitable section would be UB-356x171x67, $Z_p = 1211 \text{ cm}^3$, $M_p = 363 \text{ kNm}$

We ensure that $|M| \leq 200 \text{ kNm}$ in CE, so choose Z_p of.

$$Z_p = \frac{M_p}{\sigma_y} \geq \frac{200 \text{ k}}{300 \text{ MPa}} = 667 \times 10^6 \text{ m}^3 = 667 \text{ cm}^3$$

A suitable section would be UB-356x171x45, $Z_p = 775 \text{ cm}^3$, $M_p = 233 \text{ kNm}$

(Not a very smart design as the strength of beam was reduced at a pt. w/ pos. BM.)

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- choose two suitable UB for the spans AF and FE, [Design 2].

e now full strength $\rightarrow$ design AF as for the first design, $|M| \leq 300 \text{ kNm}$ (UB 356x171x57)

Then design FE, min. M at D, $M_D = -300 + \frac{M_2}{2} = -150 \text{ kNm}$

Now check D

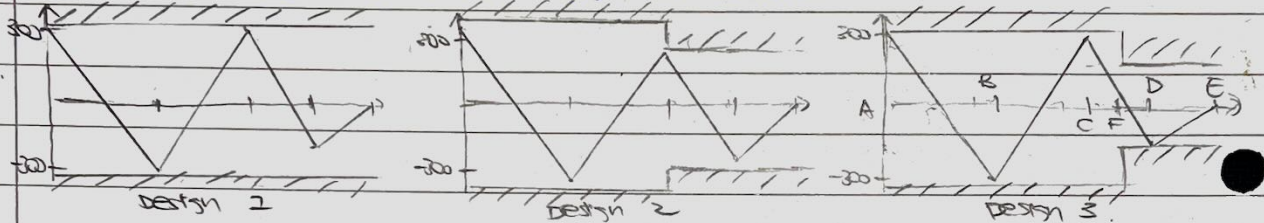
$$M_F = -\frac{300}{2} + \frac{5M_2}{4} = 75 \text{ kNm} \rightarrow \text{not limiting } \checkmark$$

$\therefore$ We ensure that $|M| \leq 150 \text{ kNm}$ in FE, so choose Z_p st.

$$Z_p = \frac{M_F}{\sigma_y} \geq \frac{150 \text{ k}}{300 \text{ M}} = 500 \times 10^{-6} \text{ m}^3 = 500 \text{ cm}^3$$

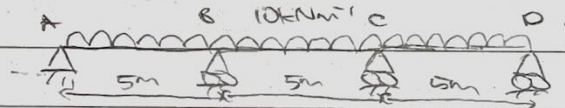
A suitable section would be UB 356x127x33, $Z_p = 542 \text{ cm}^3$, $M_p = 163 \text{ kNm}$.

- The final BM diagram for the three designs are as follows:

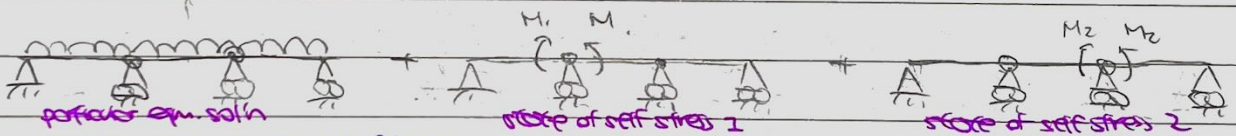


Example - three span beam

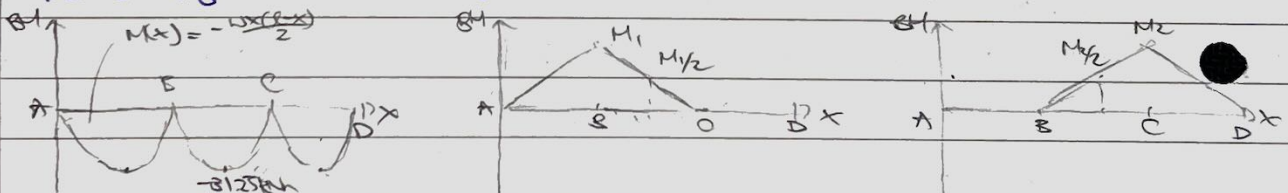
- consider the following three-span beam



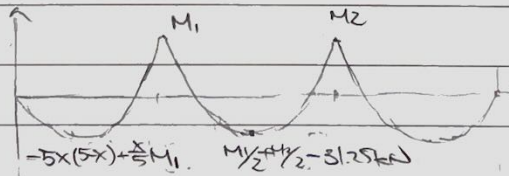
This is equivalent to acting pins and their corresponding moments.



Their BM diagrams are as follows:



The overall BM diagram is therefore



- show that $M_F = \{32; 16; 32\} \text{ kNm}$ in spans $\{AB; BC; CD\}$ is a safe design.

Design the weaker beam BC first, get $M_1 = M_2 = 16 \text{ kNm}$.

$$M(x) = 5x^2 - 25x + \frac{16}{5}x$$

check the centre of BC $M_{min} = \frac{16}{2} + \frac{16}{2} - 31.25 = -15.25 \text{ kNm} \rightarrow \text{not limiting } \checkmark$

Then design AB(CD), M_{min} when $\frac{dM}{dx} = 0$, $\frac{dM}{dx} = 10x - 21.8 = 0 \rightarrow x = 2.18 \text{ m}$

$$M_{min} = M(x=2.18) = -23.76 \text{ kNm} \rightarrow \text{not limiting } \checkmark$$

$\therefore$ We ensure that $|M| \leq 16 \text{ kNm}$ in BC, and $|M| \leq 32 \text{ kNm}$ in AB, CD.